

A Calculus of Types in Isbell Nuclei

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ABSTRACT

We identify a natural type calculus arising from a set with an associative product and a real-valued measurement. The additivity defect of the measurement defines weighted orthogonality between weighted elements; the same function is also an $\mathbb{R}$ -valued profunctor. The weighted orthogonality relation generates closed left and right types, and the paired closed types are identified with the points of the Isbell nucleus of that profunctor. This identification gives coordinates in which the product can be studied as part of the type calculus. Binary closed products need not be associative; a finite example shows that the obstruction comes from losing one of the two boundary contexts. Keeping both contexts leads to a ternary measurement. The resulting middle types carry an associative product, one-sided unit structure, and two oriented residuals, giving a noncommutative Lambek-like calculus on nuclear coordinates. In finite real coordinates, enriched Isbell nuclei have projective geometric structure, and the identification above places the type calculus in that geometric background.

1. Introduction

This paper joins two constructions with different mathematical origins. The first is the proofs-as-programs tradition of linear logic, where logical structure is read from composition and execution. The second is the enriched Isbell theory of profunctors, where a quantitative relation induces an adjunction between presheaves and copresheaves whose fixed points form a distinguished category called a nucleus. The types of the first construction and the fixed points of the second arise by different methods in different settings. This paper shows that they are the same objects, and develops the consequences of that identification.

In the first perspective, a linear realisability situation consists of a set $\mathcal{C}$, an associative execution product $\mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$, written ab , and a measurement $p: \mathcal{C} \rightarrow \mathbb{R}$. No compatibility between execution and measurement is assumed. Instead one forms a new function

$$M(a, b) = p(ab) - p(a) - p(b). \quad (1.1)$$

and declares weighted elements $(a, \alpha), (b, \beta) \in \mathcal{C} \times \mathbb{R}$ to be orthogonal when

$$\alpha + \beta \leq M(a, b).$$

Left and right types are the biorthogonally closed subsets for this relation. The types are not stipulated in advance; they are generated from execution and measurement.

This is an abstraction of a familiar theme in linear logic and realisability. In Girard's linear logic and geometry of interaction, and in the classical realisability tradition, logical formulas are controlled by orthogonality, duality, and execution. Seiller's linear realisability framework provides a general setting for this pattern. Here we work in a particularly lean real-valued case: one keeps the execution product and a numerical measurement, and then orthogonality generates the types. The simplicity of the datum $(\mathcal{C}, \text{Ex}, p)$ leaves room for noncommutative execution, for asymmetric linear implication, and for an oriented type theory adapted to settings where stricter symmetric forms of linear logic may not fit.

By viewing the set $\mathcal{C}$ as a discrete $\overline{\mathbb{R}}$ -category, the same formula $M(a, b) = p(ab) - p(a) - p(b)$ defines an $\overline{\mathbb{R}}$ -enriched profunctor. Its Isbell conjugates define adjoint functors between enriched pre and copresheaves: if $f, g : \mathcal{C} \rightarrow \overline{\mathbb{R}}$, they are given by

$$M^*f(b) = \inf_{a \in \mathcal{C}} (M(a, b) - f(a)), \quad M_*g(a) = \inf_{b \in \mathcal{C}} (M(a, b) - g(b)).$$

The nucleus $\text{Nuc}(M)$ is the fixed-point locus of this adjunction. It is a category whose objects are the pairs (f, g) with $M^*f = g$ and $M_*g = f$. Our first main result identifies the two constructions: the biorthogonally closed types of the linear realisability situation are precisely the nuclear fixed points of M . The assignments

$$A \mapsto \sup\{\alpha \in \mathbb{R} \mid (a, \alpha) \in A\}, \quad f \mapsto \{(a, \alpha) \mid \alpha \leq f(a)\},$$

implement the correspondence.

Because the two sides contribute very different structures, this alignment has many consequences. From the Isbell side, the finite real case inherits the projective metric and polyhedral geometry of the companion paper [GJST25]: witness cells, event loci, order chambers, threshold lattice towers, and the pointed gap matrix, whose entries record exactly how the metric and polyhedral structures interact. From the realisability side, the execution product contributes the oriented logical structure of insertion. A product placed inside a larger execution has a left boundary and a right boundary, and each orientation determines its own residual. From this one obtains middle types, their execution product, and a corresponding noncommutative Lambek calculus.

This oriented logical structure appears when one asks how types compose. Execution on $\mathcal{C}$ extends to weighted elements of $\mathcal{C} \times \mathbb{R}$, and hence to subsets of $\mathcal{C} \times \mathbb{R}$. After biorthogonal closure, the induced product on types need not be associative; the provided finite example below makes this failure explicit. Conceptually, one-sided binary closure does not retain both boundary contexts of a composite. The types introduced above are governed by the two-variable measurement (1.1), whereas the testing of a composite requires both a left context and a right context. This leads to the ternary measurement

$$M_3(x, b, z) = p(xbz) - p(x) - p(b) - p(z) \tag{1.2}$$

and hence to middle types defined by a middle-peripheral biorthogonality relation on $(\mathcal{C} \times \mathbb{R}) \times (\mathcal{C} \times \mathbb{R})^2$ where (b, β) is orthogonal to $((a, \alpha), (c, \gamma))$ when

$$\alpha + \beta + \gamma \leq M_3(a, b, c).$$

On middle types, the biorthogonal closure of the setwise execution product defines an associative product $\odot$. When the execution monoid has a unit, $\odot$ has a unit. It also has two residuals, $\multimap_l$ and $\multimap_r$, satisfying

$$A \odot B \subseteq C \iff B \subseteq A \multimap_l C \iff A \subseteq B \multimap_r C.$$

Together with the composition and currying laws, these operations form the noncommutative Lambek calculus carried by the linear realisability situation: its rules are derived directly from weighted orthogonality and execution.

Categorically, the middle types are the points of the nucleus of the profunctor

$$N : \mathcal{C} \multimap \mathcal{C} \times \mathcal{C}, \quad N(b, (x, z)) = M_3(x, b, z).$$

So $\odot$ defines an associative product on $\text{Nuc}(N)$. In the categorical picture, a middle type naturally creates its own measurement. A point $(f, g) \in \text{Nuc}(N)$ has copresheaf coordinate

$$g(x, z) = \inf_{b \in \mathcal{C}} (M_3(x, b, z) - f(b)).$$

For fixed middle type f , this function $g: \mathcal{C} \times \mathcal{C} \rightarrow \overline{\mathbb{R}}$ may be read as a binary profunctor $\mathcal{C} \nrightarrow \mathcal{C}$. Hence a middle type determines a new binary Isbell adjunction and a new nucleus $\text{Nuc}(g)$. These are the derived nuclei of the middle type.

The derived-nuclei theorem shows that, in the middle arrangement, the new binary nucleus produced by a middle type is controlled by the residuals of the Lambek calculus:

$$v = f \multimap_l M^*u, \quad u = f \multimap_r M_*v.$$

Moreover u and v are fixed by the binary Isbell closure operators for M . Thus the presheaf and copresheaf coordinates derived from the measurement g are genuine left and right type coordinates for the original binary realisability situation, though the pair $(u, v) \in \text{Nuc}(g)$ need not be a point of $\text{Nuc}(M)$.

The same ternary measurement M_3 has three one-variable arrangements, according to whether one singles out the left, middle, or right coordinate. The left and right arrangements of M_3 give companion derived constructions, with the same three kinds of coordinates appearing in different roles and similar derived-nuclei theorems relating the derived types to the Lambek calculus. The two-out-of-three theorem says that the three arrangements have the same pairwise intersection: compatibility with any two of them forces compatibility with the third. The common locus is the balanced locus, where every two from u, f, v is in the derived nucleus of the third, and the third can be recovered as a corresponding residual of the other two. This theorem specifies the coherence of three nuclei constructions which are related by different internal organizations of a single triple relation.

These theorems demonstrate the value of this identification of linear-realisability types with enriched Isbell nuclei. From the logical side, one has a rich noncommutative Lambek-like calculus owing to the interaction between orthogonality and execution. Yet a type, as a subset of $\mathcal{C} \times \mathbb{R}$ does not itself suggest a new binary measurement and a derived type theory. From the nuclear side, the many derived nuclei produced by factoring ternary (or higher) measurements produce a collection of derived nuclei whose internal structure and organization is hard to see, but is illuminated by the logic of types.

The identification also brings the middle product into the projective geometry of the nucleus. In the finite real case, the copresheaf coordinate of $X \odot Y$ is a convolution envelope governed by

$$M_4(a, b, c, d) = p(abcd) - p(a) - p(b) - p(c) - p(d), \quad (1.3)$$

and the corresponding product gap decomposes into input witness gaps with extra outer sharpness terms. This gives a technical bridge from the Lambek product to the geometry of execution developed in a subsequent paper.

The main results are organized as follows.

Types as binary nuclear fixed points (Theorem 28). Left and right types can be identified with the fixed points of the Isbell closure operators for the binary measurement M . Points of $\text{Nuc}(M)$ are precisely the paired types.

Peripheral products and the passage to middle types (Example 35; Proposition 42). Execution extends to paired types by taking setwise products and then biorthogonally closing. This produces the peripheral products, but an explicit example shows that they need not be associative. This leads to the introduction of middle types. On middle types, middle closure of the setwise execution product gives an associative product $\odot$.

The Lambek calculus of middle types (Propositions 52, 53, and 54). Under a unital hypothesis, the middle product has a unit, and it carries left and right residuals $\multimap_l, \multimap_r$. The residuation, currying, composition, and partial transitivity laws give the noncommutative

Lambek calculus naturally associated with execution. The rules are derived directly from weighted orthogonality and execution.

Internalization of derived nuclei (Theorem 58). Middle types are identified with the points of $\text{Nuc}(N)$, where N is a profunctor defined by the ternary measurement M_3 . For a middle nuclear point $(f, g) \in \text{Nuc}(N)$, the copresheaf coordinate g defines a binary measurement $\mathcal{C} \rightarrow \mathcal{C}$ and hence has its own nucleus. Every point $(u, v) \in \text{Nuc}(g)$ satisfies

$$v = f \multimap_l M^*u, \quad u = f \multimap_r M_*v,$$

and the same theorem gives explicit duals placing u in $\text{im } M_*$ and v in $\text{im } M^*$. Derived nuclei therefore return to the binary fixed-coordinate loci of M , and their coordinates are computed by the residuals of the Lambek calculus.

Two-out-of-three coherence (Theorem 62). The left, middle, and right arrangements of M_3 give three fixed-point conditions on triples (u, f, v) . Their pairwise intersections agree with their triple intersection. Equivalently, compatibility with any two arrangements forces compatibility with the third, and the common locus is the balanced locus where each coordinate is recovered as the residual of the other two.

Product envelopes and factor gaps (Corollary 63 and Proposition 64). In the finite real-valued situation, the middle product descends to a well-defined operation on $\mathbb{P}\text{Nuc}(N)$, and its copresheaf is the convolution envelope obtained from M_4 . The factor-gap identity decomposes a candidate product witness into shifted input gap terms together with the sharpness conditions for the two outer minimizations; in particular, an envelope-computing factorisation forces all four quantities to vanish.

Examples. Throughout the paper we use explicit finite examples to show that the main results are sharp. A single four-element execution monoid is small enough to be computed directly and illustrates a number of phenomena: the left closed product need not be associative; the two-out-of-three hypothesis in the coherence theorem cannot be weakened to a single arrangement; and the left, middle, and right units in the Lambek calculus are distinct, with the middle unit profile strictly below the pointwise minimum of the one-sided unit profiles.

Relation to prior work

The linear realisability framework used here belongs to the lineage of Girard’s linear logic and geometry of interaction [Gir87, Gir11], ludics [Gir01], and the classical realisability tradition of Kleene, Kreisel, Krivine, and van Oosten [Kle45, Kre59, Kri09, vO08]. Seiller introduced linear realisability [Sei24] to isolate the common structure of geometry-of-interaction models and their extensions [Sei12, Sei16]. The Lambek calculus originates in Lambek’s work on the mathematics of sentence structure [Lam58]; in the present paper its noncommutative residuated structure is derived from weighted execution and middle orthogonality.

The Isbell nucleus originates in Isbell’s work on adequate subcategories [Isb60]. Modern treatments of Isbell duality and enriched nuclei include Avery–Leinster [AL21] and Willerton’s work in the metric setting [Wil13, Wil14, Wil15], with the enriched metric viewpoint going back to Lawvere [Law73] and Kelly [Kel82]. Jarvis studies monoidal structures on nuclei of profunctors in a compatible monoidal setting [Jar25].

The geometric results used here are developed in the companion paper [GJST25]. There the projective nucleus of a finite real profunctor is studied through its gap matrix, witness cells, event loci, order chambers, formal concept lattice towers, and Chebyshev centers. The enrichment viewpoint also appears in work on tropical convexity, directed metrics, and language-like structures, including [DS04, Ell17, Fuj19, GV24, BTv22, BGT24, BV25].

Organization. Section 2 establishes the $\overline{\mathbb{R}}$ -enriched conventions and recalls Isbell duality for profunctors. Section 3 introduces linear realisability situations, weighted orthogonality, and types. Section 4 proves the nucleus–types correspondence. Section 5 explains how execution extends to peripheral types and analyzes the failure of associativity there. Section 6 introduces middle types and proves associativity of the middle product. Section 7 develops units, residuals, and the Lambek-style calculus of linear arrows. Section 8 proves the internalization theorem for derived nuclei, the corrected convolution formulas for middle products, projective well-definedness of the product, and the ternary coherence theorem. Section 9 treats the finite real case and records the product envelope and factor-gap identities connecting $\odot$ with the witness geometry of the middle nucleus.

2. Isbell duality over the extended reals

This section fixes the enriched conventions needed later to compare two kinds of closure. On the logical side, types will be closed subsets for a weighted orthogonality relation. On the Isbell side, the same objects will appear as fixed points of closure operators associated to a profunctor. These conventions provide the arithmetic, profunctor, and Isbell-nucleus language for the binary correspondence and for the derived nuclei of Section 8.

2.1. The arithmetic of $\overline{\mathbb{R}}$

Let $\overline{\mathbb{R}} = [-\infty, +\infty]$ regarded as a poset category where a morphism $x \rightarrow y$ exists precisely when $x \leq y$, and arbitrary limits and colimits are infima and suprema. The monoidal structure is the extension of addition on $\mathbb{R}$ used throughout this paper. Its unit is 0, and the endpoint convention is

$$-\infty + y = -\infty \quad \text{for all } y \in \overline{\mathbb{R}},$$

including $y = +\infty$. This is not an incidental convention. We use $\overline{\mathbb{R}}$ as a closed ordered monoidal base: for every $y \in \overline{\mathbb{R}}$, translation $x \mapsto x + y$ is required to preserve arbitrary suprema and hence to have a right adjoint. In particular it preserves the bottom element, forcing $-\infty + y = -\infty$. We write this right adjoint as $[y, -]$, or more suggestively as subtraction:

$$x + y \leq z \iff x \leq [y, z] = z - y.$$

Equivalently,

$$z - y = \sup\{x \in \overline{\mathbb{R}} \mid x + y \leq z\}. \quad (2.1)$$

For finite real numbers this is ordinary subtraction. At the endpoints it is internal-hom notation; for example $-\infty - (-\infty) = +\infty$, while $-\infty + (+\infty) = -\infty$.

The use of both endpoints is essential for the type story. A subset $A \subseteq C \times \mathbb{R}$ determines a function

$$c \mapsto \sup\{\alpha \in \mathbb{R} \mid (c, \alpha) \in A\}.$$

The value is $-\infty$ over an empty fibre and may be $+\infty$ over a fibre containing arbitrarily large real weights. Thus the Isbell coordinates of types naturally live in $\overline{\mathbb{R}}$, even when the measurement function in the realisability situation is real-valued.

REMARK 1 Endpoint convention. All displayed expressions of the form $z - y$ in this section mean the residual (2.1). Endpoint manipulations in later sections, especially those involving $\Omega_f \subseteq C \times \mathbb{R}$ rather than $C \times \overline{\mathbb{R}}$, are read with this convention.

2.2. Categories, presheaves, and profunctors

DEFINITION 2. A small $\overline{\mathbb{R}}$ -category $\mathcal{C}$ consists of a set $\text{Ob}(\mathcal{C})$ and hom-values $\mathcal{C}(c, c') \in \overline{\mathbb{R}}$ satisfying, for all c, c', c'' ,

$$0 \leq \mathcal{C}(c, c), \quad (2.2a)$$

$$\mathcal{C}(c, c') + \mathcal{C}(c', c'') \leq \mathcal{C}(c, c''). \quad (2.2b)$$

We call $\mathcal{C}$ *normalized* if $\mathcal{C}(c, c) = 0$ for every object c .

The base $\overline{\mathbb{R}}$ is itself an $\overline{\mathbb{R}}$ -category, with hom-value

$$\overline{\mathbb{R}}(x, y) = y - x,$$

where the right-hand side is the residual (2.1). The opposite $\mathcal{C}^{\text{op}}$ has the same objects as $\mathcal{C}$ and hom-values $\mathcal{C}^{\text{op}}(c, c') = \mathcal{C}(c', c)$.

DEFINITION 3. An $\overline{\mathbb{R}}$ -functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is a function on objects such that

$$\mathcal{C}(c, c') \leq \mathcal{D}(Fc, Fc')$$

for all objects c, c' of $\mathcal{C}$.

For $\overline{\mathbb{R}}$ -categories $\mathcal{A}$ and $\mathcal{B}$, we write $[\mathcal{A}, \mathcal{B}]$ for the $\overline{\mathbb{R}}$ -category of $\overline{\mathbb{R}}$ -functors $\mathcal{A} \rightarrow \mathcal{B}$, where hom-values are computed pointwise: if $F, G : \mathcal{A} \rightarrow \mathcal{B}$ are two such functors, then

$$[\mathcal{A}, \mathcal{B}](F, G) = \inf_{a \in \mathcal{A}} \mathcal{B}(Fa, Ga). \quad (2.3)$$

When the ambient functor category is clear, we write $[F, G]$ for this hom-value.

A *presheaf* on $\mathcal{C}$ is an $\overline{\mathbb{R}}$ -functor $f : \mathcal{C}^{\text{op}} \rightarrow \overline{\mathbb{R}}$. A *copresheaf* on $\mathcal{D}$ is an $\overline{\mathbb{R}}$ -functor $g : \mathcal{D} \rightarrow \overline{\mathbb{R}}$. In the Isbell adjunction below, copresheaves are regarded as objects of the opposite enriched category $[\mathcal{D}, \overline{\mathbb{R}}]^{\text{op}}$. Thus the underlying functions are still $g : \mathcal{D} \rightarrow \overline{\mathbb{R}}$, but the enriched order is reversed. For presheaves $f, f' : \mathcal{C}^{\text{op}} \rightarrow \overline{\mathbb{R}}$ and copresheaves $g, g' : \mathcal{D} \rightarrow \overline{\mathbb{R}}$, the hom-values are

$$\begin{aligned} [f, f'] &= \inf_{c \in \mathcal{C}} (f'(c) - f(c)), \\ [g, g']_{[\mathcal{D}, \overline{\mathbb{R}}]^{\text{op}}} &= [\mathcal{D}, \overline{\mathbb{R}}](g', g) = \inf_{d \in \mathcal{D}} (g(d) - g'(d)). \end{aligned}$$

Consequently the underlying order on presheaves is pointwise order, while the underlying order on copresheaves in $[\mathcal{D}, \overline{\mathbb{R}}]^{\text{op}}$ is the opposite of pointwise order.

REMARK 4 Variance convention. Unless explicitly stated otherwise, inequalities between displayed presheaves and copresheaves are pointwise inequalities. This agrees with the enriched order for presheaves, but not for copresheaves viewed in $[\mathcal{D}, \overline{\mathbb{R}}]^{\text{op}}$.

The enriched Yoneda lemma gives the formulas

$$[\mathcal{C}(-, c), f] = f(c), \quad [g, \mathcal{D}(d, -)]_{[\mathcal{D}, \overline{\mathbb{R}}]^{\text{op}}} = g(d).$$

These formulas fix the representable presheaves and copresheaves that occur when a profunctor is evaluated in one variable. The tensor product of $\overline{\mathbb{R}}$ -categories $\mathcal{C}$ and $\mathcal{D}$ is the $\overline{\mathbb{R}}$ -category $\mathcal{C} \otimes \mathcal{D}$ whose objects are pairs (c, d) and whose hom-values are

$$(\mathcal{C} \otimes \mathcal{D})((c, d), (c', d')) = \mathcal{C}(c, c') + \mathcal{D}(d, d').$$

DEFINITION 5. A *profunctor* $M : \mathcal{C} \nrightarrow \mathcal{D}$ is an $\overline{\mathbb{R}}$ -functor

$$M : \mathcal{C}^{\text{op}} \otimes \mathcal{D} \rightarrow \overline{\mathbb{R}}.$$

Equivalently, it is a function $M : \text{Ob}(\mathcal{C}) \times \text{Ob}(\mathcal{D}) \rightarrow \overline{\mathbb{R}}$ such that, for all $c, c' \in \mathcal{C}$ and $d, d' \in \mathcal{D}$,

$$\mathcal{C}(c', c) + \mathcal{D}(d, d') \leq [M(c, d), M(c', d')] = M(c', d') - M(c, d). \quad (2.4)$$

A set S will often be viewed as a discrete $\overline{\mathbb{R}}$ -category, again denoted S , by

$$S(s, s') = \begin{cases} 0 & s = s', \\ -\infty & s \neq s'. \end{cases}$$

For discrete $\mathcal{C}$ and $\mathcal{D}$, the profunctor condition (2.4) imposes no restriction: every function $M : \mathcal{C} \times \mathcal{D} \rightarrow \overline{\mathbb{R}}$ is a profunctor. This is the case when the binary measurement

$$M(a, b) = p(ab) - p(a) - p(b)$$

is regarded as an $\overline{\mathbb{R}}$ -profunctor.

2.3. Isbell duality and the nucleus

Let $M : \mathcal{C} \nrightarrow \mathcal{D}$ be a profunctor. For each $d \in \mathcal{D}$, the column $M(-, d)$ is a presheaf on $\mathcal{C}$; for each $c \in \mathcal{C}$, the row $M(c, -)$ is a copresheaf on $\mathcal{D}$. The Isbell conjugates extend these assignments from rows and columns to arbitrary presheaves and copresheaves.

DEFINITION 6. Define maps

$$\begin{aligned} M^* : [\mathcal{C}^{\text{op}}, \overline{\mathbb{R}}] &\rightarrow [\mathcal{D}, \overline{\mathbb{R}}]^{\text{op}}, \\ M_* : [\mathcal{D}, \overline{\mathbb{R}}]^{\text{op}} &\rightarrow [\mathcal{C}^{\text{op}}, \overline{\mathbb{R}}] \end{aligned}$$

by

$$(M^*f)(d) := \inf_{c \in \mathcal{C}} (M(c, d) - f(c)), \quad (2.5)$$

$$(M_*g)(c) := \inf_{d \in \mathcal{D}} (M(c, d) - g(d)). \quad (2.6)$$

The maps M^* and M_* are the Isbell conjugates induced by M .

All minus signs in (2.5)–(2.6) are residuals in $\overline{\mathbb{R}}$.

PROPOSITION 7. The assignments (2.5)–(2.6) define $\overline{\mathbb{R}}$ -functors and satisfy

$$M^* \dashv M_*,$$

meaning that

$$[\mathcal{D}, \overline{\mathbb{R}}]^{\text{op}}(M^*f, g) = [\mathcal{C}^{\text{op}}, \overline{\mathbb{R}}](f, M_*g)$$

for all $f \in [\mathcal{C}^{\text{op}}, \overline{\mathbb{R}}]$ and $g \in [\mathcal{D}, \overline{\mathbb{R}}]^{\text{op}}$.

Proof. We record the verification in the present $\overline{\mathbb{R}}$ -valued notation. The functoriality calculation is the usual one for Isbell conjugacy. If $\delta = [f, f'] = \inf_c (f'(c) - f(c))$, then $f(c) + \delta \leq f'(c)$ for every c . By the residuation law, this implies

$$\delta + (M(c, d) - f'(c)) \leq M(c, d) - f(c).$$

Since $(M^*f')(d) \leq M(c, d) - f'(c)$ for every c , we obtain

$$\delta + (M^*f')(d) \leq M(c, d) - f(c)$$

for every c , and hence $\delta + (M^*f')(d) \leq (M^*f)(d)$. This is precisely the enriched functoriality inequality for M^* . The proof for M_* is the same. For the adjunction, residuation by $g(d)$ preserves infima, since it is a right adjoint. Hence

$$\begin{aligned} [\mathcal{D}, \overline{\mathbb{R}}]^{\text{op}}(M^*f, g) &= \inf_d \left(\inf_c (M(c, d) - f(c)) - g(d) \right) \\ &= \inf_d \inf_c (M(c, d) - f(c) - g(d)) \\ &= \inf_c \left(\inf_d (M(c, d) - g(d)) - f(c) \right) \\ &= [\mathcal{C}^{\text{op}}, \overline{\mathbb{R}}](f, M_*g). \end{aligned}$$

□

REMARK 8 Variance convention for Isbell conjugacy. The displayed computation fixes the variance convention for the Isbell adjunction used below.

LEMMA 9. *With respect to pointwise order, the maps M^* and M_* are order-reversing:*

$$f \leq f' \Rightarrow M^*f' \leq M^*f, \quad g \leq g' \Rightarrow M_*g' \leq M_*g.$$

Proof. If $f \leq f'$, then $M(c, d) - f'(c) \leq M(c, d) - f(c)$ for all c, d , because the residual is antitone in its second variable. Taking infima gives $M^*f' \leq M^*f$. The proof for M_* is identical. □

Thus the composites

$$\begin{aligned} \text{cl}_{\mathcal{C}} &:= M_*M^*: [\mathcal{C}^{\text{op}}, \overline{\mathbb{R}}] \rightarrow [\mathcal{C}^{\text{op}}, \overline{\mathbb{R}}], \\ \text{cl}_{\mathcal{D}} &:= M^*M_*: [\mathcal{D}, \overline{\mathbb{R}}]^{\text{op}} \rightarrow [\mathcal{D}, \overline{\mathbb{R}}]^{\text{op}} \end{aligned}$$

are monotone for pointwise order. The adjunction $M^* \dashv M_*$ implies that they are closure operators:

$$\begin{aligned} f &\leq \text{cl}_{\mathcal{C}}(f), & \text{cl}_{\mathcal{C}}^2 &= \text{cl}_{\mathcal{C}}, \\ g &\leq \text{cl}_{\mathcal{D}}(g), & \text{cl}_{\mathcal{D}}^2 &= \text{cl}_{\mathcal{D}}. \end{aligned} \tag{2.7}$$

DEFINITION 10. The *nucleus* of M is the $\overline{\mathbb{R}}$ -category $\text{Nuc}(M)$ whose objects are pairs

$$\text{Nuc}(M) = \{ (f, g) \mid f \in [\mathcal{C}^{\text{op}}, \overline{\mathbb{R}}], g \in [\mathcal{D}, \overline{\mathbb{R}}]^{\text{op}}, g = M^*f, f = M_*g \},$$

with hom-values inherited from either side:

$$\text{Nuc}(M)((f, g), (f', g')) = [\mathcal{C}^{\text{op}}, \overline{\mathbb{R}}](f, f') = [\mathcal{D}, \overline{\mathbb{R}}]^{\text{op}}(g, g').$$

The equality follows from the adjunction $M^* \dashv M_*$.

The fixed-point description of the nucleus is the form used throughout the rest of the paper.

PROPOSITION 11. *There are canonical isomorphisms of $\overline{\mathbb{R}}$ -categories*

$$\text{Nuc}(M) \cong \text{Fix}(\text{cl}_{\mathcal{C}}) \cong \text{Fix}(\text{cl}_{\mathcal{D}}) \cong \text{im}(M^*) \cong \text{im}(M_*),$$

where $\text{Fix}(\text{cl}_{\mathcal{C}}) = \{f \mid \text{cl}_{\mathcal{C}}(f) = f\}$ and similarly for $\mathcal{D}$, and $\text{im}(M^*)$ and $\text{im}(M_*)$ denote the full subcategories spanned by objects of the corresponding forms.

Proof. The projection $(f, g) \mapsto f$ identifies $\text{Nuc}(M)$ with $\text{Fix}(M_*M^*)$: indeed, $(f, g) \in \text{Nuc}(M)$ if and only if $f = M_*g = M_*M^*f$. The projection $(f, g) \mapsto g$ gives the dual identification with $\text{Fix}(M^*M_*)$. For any closure operator, fixed points and image agree. Finally, M_* already lands in $\text{Fix}(M_*M^*)$, because the expansion $g \leq M^*M_*g$ and antitonicity of M_* give $M_*M^*M_*g \leq M_*g$, while expansion gives the reverse inequality. Thus $M_*M^*M_* = M_*$. Dually, $M^*M_*M^* = M^*$. $\square$

REMARK 12. Proposition 11 gives an explicit way to produce objects of $\text{Nuc}(M)$. For any presheaf f , the pair $(\text{cl}_{\mathcal{C}}(f), M^*f)$ lies in $\text{Nuc}(M)$. For any copresheaf g , the pair $(M_*g, \text{cl}_{\mathcal{D}}(g))$ lies in $\text{Nuc}(M)$.

REMARK 13. A presheaf f is $\text{cl}_{\mathcal{C}}$ -closed if and only if it is the largest presheaf, for the pointwise order, among those with the same M^* -image:

$$M^*h = M^*f \implies h \leq f.$$

Dually, a copresheaf g is $\text{cl}_{\mathcal{D}}$ -closed if and only if it is the largest copresheaf, for the pointwise order, among those with the same M_* -image.

In the next section, we introduce a real linear realisability situation: a set $\mathcal{C}$ equipped with an associative execution operation and a real-valued measurement. From these data one obtains a binary measurement $M(a, b) = p(ab) - p(a) - p(b)$; after viewing the set $\mathcal{C}$ as a discrete $\overline{\mathbb{R}}$ -category, this measurement becomes an $\overline{\mathbb{R}}$ -profunctor of the kind studied above. The Isbell-nuclear formalism then provides one side of the comparison with the logical types generated by orthogonality.

3. Linear realisability and types

Section 2 fixed the enriched side of the comparison: a profunctor M determines conjugate Isbell closure operators and a nucleus. We now construct the logical side. A linear realisability situation supplies an execution product and a real-valued function; from these data we obtain a binary measurement $M(a, b)$, a weighted orthogonality relation on $\mathcal{C} \times \mathbb{R}$, and the left and right types generated by the associated orthogonal-complement operations.

3.1. The realisability datum and its measurement

DEFINITION 14. A *real linear realisability situation* is a triple $(\mathcal{C}, \text{Ex}, p)$, where $\mathcal{C}$ is a set, $\text{Ex}: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ is an associative execution product, and $p: \mathcal{C} \rightarrow \mathbb{R}$ is a function. We write

$$\text{Ex}(a, b) = ab.$$

No unit is assumed, and no compatibility between p and execution is assumed. Instead we measure the defect of execution by the function

$$M: \mathcal{C} \times \mathcal{C} \longrightarrow \mathbb{R}, \quad M(a, b) := p(ab) - p(a) - p(b).$$

Seiller's linear-realisation framework uses double-bracket notation for an abstract measurement; see, for example, Seiller's habilitation thesis [Sei24, Definition 10.1.1 and Remark 10.1.1].

In the real-valued situation considered here, the measurement is the special coboundary $M(a, b) = p(ab) - p(a) - p(b)$, and we write it as M to emphasize its role as the profunctor to which the Isbell formalism of Section 2 applies.

PROPOSITION 15 Trefoil identity. *The measurement M satisfies*

$$M(ab, c) + M(a, b) = M(a, bc) + M(b, c) \quad (3.1)$$

for all $a, b, c \in \mathcal{C}$.

Proof. Associativity gives $p((ab)c) = p(a(bc)) = p(abc)$. Therefore

$$\begin{aligned} M(ab, c) + M(a, b) &= p(abc) - p(ab) - p(c) + p(ab) - p(a) - p(b) \\ &= p(abc) - p(a) - p(b) - p(c) \\ &= p(abc) - p(a) - p(bc) + p(bc) - p(b) - p(c) \\ &= M(a, bc) + M(b, c). \end{aligned}$$

□

REMARK 16. With the signed bar convention

$$(\delta p)(a, b) = p(ab) - p(a) - p(b),$$

we have $M = \delta p$. For a two-cochain Q , the corresponding convention is

$$(\delta Q)(a, b, c) = Q(ab, c) - Q(a, bc) + Q(a, b) - Q(b, c).$$

Thus the trefoil identity is the cocycle equation $\delta M = 0$, equivalently $\delta^2 p = 0$. In what follows we use the displayed trefoil identity rather than the bar-complex terminology.

When $\mathcal{C}$ is regarded as a discrete $\overline{\mathbb{R}}$ -category, the real-valued function M is also an $\overline{\mathbb{R}}$ -valued profunctor by inclusion $\mathbb{R} \subset \overline{\mathbb{R}}$. This observation is the bridge to the Isbell constructions of Section 2.

3.2. Weighted orthogonality

The type construction is generated by a weighted form of orthogonality. For $(a, \alpha), (b, \beta) \in \mathcal{C} \times \mathbb{R}$, define

$$(a, \alpha) \perp (b, \beta) \iff \alpha + \beta \leq M(a, b). \quad (3.2)$$

The unweighted inequality $0 \leq M(a, b)$ may be used as a sign test for interaction between a and b , but the formal objects of this paper are the weighted subsets of $\mathcal{C} \times \mathbb{R}$ closed under (3.2). For later multiplicative constructions, it is useful to record the associated weighted execution product

$$(a, \alpha)(b, \beta) := (ab, \alpha + \beta - M(a, b)).$$

Its associativity is exactly the trefoil identity: the two second coordinates of $((a, \alpha)(b, \beta))(c, \gamma)$ and $(a, \alpha)((b, \beta)(c, \gamma))$ are

$$\alpha + \beta + \gamma - M(a, b) - M(ab, c) \quad \text{and} \quad \alpha + \beta + \gamma - M(b, c) - M(a, bc),$$

which agree by (3.1). The complement calculus below uses only the relation $\perp$; later sections use the displayed product when discussing products of types.

3.3. Complements and types

For subsets $A, B \subseteq \mathcal{C} \times \mathbb{R}$, define the left and right orthogonal complements by

$${}^\perp B := \{(a, \alpha) \mid \alpha + \beta \leq M(a, b) \text{ for all } (b, \beta) \in B\},$$

and

$$A^\perp := \{(b, \beta) \mid \alpha + \beta \leq M(a, b) \text{ for all } (a, \alpha) \in A\}.$$

The two constructions are oriented. Since $M(a, b)$ need not equal $M(b, a)$, the left and right complements need not agree.

DEFINITION 17. For $A \subseteq \mathcal{C} \times \mathbb{R}$, its *left closure* is

$$\text{cl}_L(A) := {}^\perp(A^\perp).$$

Its *right closure* is

$$\text{cl}_R(A) := ({}^\perp A)^\perp.$$

DEFINITION 18. A subset $A \subseteq \mathcal{C} \times \mathbb{R}$ is a *left type* if $A = {}^\perp B$ for some $B \subseteq \mathcal{C} \times \mathbb{R}$. A subset $B \subseteq \mathcal{C} \times \mathbb{R}$ is a *right type* if $B = A^\perp$ for some $A \subseteq \mathcal{C} \times \mathbb{R}$. A *paired type* is a pair (A, B) such that

$$A = {}^\perp B \quad \text{and} \quad B = A^\perp.$$

Thus left and right types are not primitive data. They are the closed objects generated by weighted orthogonality. We now describe the properties of our orthogonal-complement operation.

PROPOSITION 19 Orthogonal-complement calculus. *Let all sets below be subsets of $\mathcal{C} \times \mathbb{R}$.*

- (i) *If $A_1 \subseteq A_2$, then $A_2^\perp \subseteq A_1^\perp$. If $B_1 \subseteq B_2$, then ${}^\perp B_2 \subseteq {}^\perp B_1$.*
- (ii) *For every A and B ,*

$$A \subseteq {}^\perp(A^\perp), \quad B \subseteq ({}^\perp B)^\perp.$$

- (iii) *Alternating three complements cancels to one complement:*

$${}^\perp({}^\perp({}^\perp B)^\perp) = {}^\perp B, \quad ({}^\perp(A^\perp))^\perp = A^\perp.$$

- (iv) *A subset A is a left type if and only if $\text{cl}_L(A) = A$. A subset B is a right type if and only if $\text{cl}_R(B) = B$. In particular, $\text{cl}_L(A)$ is a left type and $\text{cl}_R(B)$ is a right type for arbitrary A and B .*
- (v) *The assignments*

$$A \mapsto A^\perp, \quad B \mapsto {}^\perp B$$

restrict to inverse bijections between left types and right types. Under these bijections, a left type A and a right type B correspond precisely when (A, B) is a paired type.

Proof. The first assertion is immediate from the definitions: enlarging the set to be tested against imposes more inequalities. For the second assertion, if $(a, \alpha) \in A$ and $(b, \beta) \in A^\perp$, then $(a, \alpha) \perp (b, \beta)$ by definition of $A^\perp$. Hence $(a, \alpha) \in {}^\perp(A^\perp)$. The proof of $B \subseteq ({}^\perp B)^\perp$ is the same with left and right interchanged.

For the first identity in (iii), apply (ii) to the set ${}^\perp B$ to get

$${}^\perp B \subseteq {}^\perp({}^\perp({}^\perp B)^\perp).$$

On the other hand, (ii) gives $B \subseteq (\perp B)^\perp$, and antitonicity of the left complement gives

$$\perp((\perp B)^\perp) \subseteq \perp B.$$

The second identity in (iii) is analogous. If A is a left type, say $A = \perp B$, then (iii) gives

$$\text{cl}_L(A) = \perp(A^\perp) = \perp((\perp B)^\perp) = \perp B = A.$$

Conversely, if $\text{cl}_L(A) = A$, then $A = \perp(A^\perp)$, so A is a left type. The right-type assertion is parallel. The final assertion then follows directly: if A is a left type, then $A = \perp(A^\perp)$, so A is paired with $A^\perp$; if B is a right type, then $B = (\perp B)^\perp$, so B is paired with $\perp B$. $\square$

We write

$$\text{LType}(M), \quad \text{RType}(M), \quad \text{PType}(M)$$

for the left types, right types, and paired types determined by M . The preceding proposition gives canonical bijections among these three presentations:

$$\begin{array}{ccc} & \text{PType}(M) & \\ (A,B) \mapsto B \swarrow & & \searrow (A,B) \mapsto A \\ \text{RType}(M) & \xrightleftharpoons[B \mapsto \perp B]{A \mapsto A^\perp} & \text{LType}(M) \end{array}$$

REMARK 20 Endpoint conventions and profile recovery. This section constructs types as subsets of $\mathcal{C} \times \mathbb{R}$. The comparison with Isbell fixed points in Section 4 passes through $\overline{\mathbb{R}}$ -valued presheaf coordinates. In that comparison one must distinguish arbitrary fibres from the lower rays in the weight coordinate that are recovered from their suprema; this is an additional extended-real endpoint convention, not a consequence of the double-complement calculus alone.

The enriched language developed in Section 2 now meets the construction from weighted orthogonality developed here. In the next section we introduce the maps Ω and φ , which translate between weighted subsets of $\mathcal{C} \times \mathbb{R}$ and $\overline{\mathbb{R}}$ -valued coordinates, and use them to compare the left and right types above with fixed points in the Isbell nucleus of M .

4. The nucleus-types correspondence

Section 2 associates to a profunctor M an Isbell adjunction and its nucleus. Section 3 associates to the same measurement, when it comes from a real linear realisability situation, a weighted orthogonality relation on $\mathcal{C} \times \mathbb{R}$, together with its left and right complement operations. We now compare these two closure constructions.

Throughout this section $\mathcal{C}$ is the set from Section 3, viewed as a discrete $\overline{\mathbb{R}}$ -category, and

$$M(a, b) = p(ab) - p(a) - p(b)$$

is regarded as an $\overline{\mathbb{R}}$ -valued profunctor by the inclusion $\mathbb{R} \subset \overline{\mathbb{R}}$. Since $\mathcal{C}$ is discrete, presheaf and copresheaf coordinates are simply functions $\mathcal{C} \rightarrow \overline{\mathbb{R}}$; the distinction remains one of variance, as in Section 2.

4.1. Profiles and recovered weighted subsets

For a function $f : \mathcal{C} \rightarrow \overline{\mathbb{R}}$, define the weighted subset

$$\Omega_f := \{(a, \alpha) \in \mathcal{C} \times \mathbb{R} \mid \alpha \leq f(a)\}.$$

Thus an $+\infty$ -value gives the whole real fibre, while a $-\infty$ -value gives the empty fibre. Notice that Ω_f is always a subset of $\mathcal{C} \times \mathbb{R}$, not of $\mathcal{C} \times \overline{\mathbb{R}}$. Conversely, any weighted subset $X \subseteq \mathcal{C} \times \mathbb{R}$ has a profile

$$\varphi_X(c) := \sup\{\xi \in \mathbb{R} \mid (c, \xi) \in X\}, \quad \sup \emptyset = -\infty.$$

For a subset used in the left variable we write

$$\lambda_A := \varphi_A,$$

and for a subset used in the right variable we write

$$\rho_B := \varphi_B.$$

This separates the two coordinates of the Isbell adjunction: λ_A is a left, or presheaf, coordinate and ρ_B is a right, or copreheaf, coordinate.

DEFINITION 21. A subset $X \subseteq \mathcal{C} \times \mathbb{R}$ is *recovered from its profile* if

$$X = \Omega_{\varphi_X}.$$

Equivalently, each fibre of X is either empty, a closed lower ray $(-\infty, r]$ with $r \in \mathbb{R}$, or all of $\mathbb{R}$. This is the form, with extended-real endpoint conventions, of the familiar condition that the subset is down-closed and contains the supremum of each finite fibre.

LEMMA 22. For every $f : \mathcal{C} \rightarrow \overline{\mathbb{R}}$,

$$\varphi_{\Omega_f} = f.$$

For every $X \subseteq \mathcal{C} \times \mathbb{R}$, one has

$$X \subseteq \Omega_{\varphi_X},$$

with equality if and only if X is recovered from its profile.

Proof. Fix $a \in \mathcal{C}$. If $f(a) \in \mathbb{R}$, then the fibre of Ω_f over a is $(-\infty, f(a)]$, whose supremum is $f(a)$. If $f(a) = +\infty$, the fibre is all of $\mathbb{R}$, whose supremum is $+\infty$. If $f(a) = -\infty$, the fibre is empty, whose supremum is $-\infty$ by convention. Hence $\varphi_{\Omega_f} = f$. If $(c, \xi) \in X$, then $\xi \leq \varphi_X(c)$ by definition of the supremum, so $X \subseteq \Omega_{\varphi_X}$. The final assertion is just the condition $X = \Omega_{\varphi_X}$. $\square$

REMARK 23. The profile does not recover an arbitrary fibre. For example, over a single object, the subsets $\{0\}$, $(-\infty, 0)$, and $(-\infty, 0]$ all have profile 0, but only $(-\infty, 0]$ is recovered from its profile. Thus the correspondence below is a statement about subsets recovered from their profiles, and in particular about types, not about arbitrary weighted subsets.

4.2. Orthogonality in profile coordinates

Recall from Section 2 that the Isbell conjugates associated to M are

$$(M^*f)(b) = \inf_{a \in \mathcal{C}} (M(a, b) - f(a)), \quad (M_*g)(a) = \inf_{b \in \mathcal{C}} (M(a, b) - g(b)),$$

where subtraction means the residual in $\overline{\mathbb{R}}$. The first formula sends a left coordinate to a right coordinate; the second sends a right coordinate to a left coordinate.

PROPOSITION 24. *For all $A, B \subseteq \mathcal{C} \times \mathbb{R}$,*

$$A^\perp = \Omega_{M^* \lambda_A}, \quad {}^\perp B = \Omega_{M_* \rho_B}.$$

Proof. We prove the first identity. Let $(b, \beta) \in \mathcal{C} \times \mathbb{R}$. By definition of the right complement,

$$(b, \beta) \in A^\perp \iff \alpha + \beta \leq M(a, b) \text{ for all } (a, \alpha) \in A.$$

Fixing a , this condition over the fibre of A is equivalent to

$$\lambda_A(a) + \beta \leq M(a, b),$$

because translation by the real number β preserves suprema in $\overline{\mathbb{R}}$. This remains meaningful at the endpoints: an empty fibre gives $\lambda_A(a) = -\infty$ and no constraint, while an unbounded fibre gives $\lambda_A(a) = +\infty$, which in the present real-valued measurement setting gives no real β satisfying the inequality. By residuation, the displayed inequality is equivalent to

$$\beta \leq M(a, b) - \lambda_A(a).$$

Requiring this for every a is equivalent to

$$\beta \leq \inf_{a \in \mathcal{C}} (M(a, b) - \lambda_A(a)) = (M^* \lambda_A)(b).$$

This is precisely $(b, \beta) \in \Omega_{M^* \lambda_A}$. The proof of ${}^\perp B = \Omega_{M_* \rho_B}$ is the same calculation with the two variables interchanged. $\square$

COROLLARY 25. *Every left type and every right type is recovered from its profile. More precisely, if A is a left type and B is a right type, then*

$$A = \Omega_{\lambda_A}, \quad B = \Omega_{\rho_B}.$$

Proof. A right type B has the form $X^\perp$ for some X , and Proposition 24 writes it as $\Omega_{M^* \lambda_X}$. A left type A has the form ${}^\perp Y$ and is handled by the second identity in the same proposition. The identities with λ_A and ρ_B then follow from Lemma 22. $\square$

4.3. Fixed points and paired types

The preceding proposition identifies orthogonal complement with Isbell conjugacy. Applying it twice identifies biorthogonal closure with the two Isbell closure operators.

COROLLARY 26. *Let $f : \mathcal{C} \rightarrow \overline{\mathbb{R}}$ be a left coordinate and $g : \mathcal{C} \rightarrow \overline{\mathbb{R}}$ a right coordinate. Then*

$$\Omega_f \text{ is a left type} \iff M_* M^* f = f,$$

and

$$\Omega_g \text{ is a right type} \iff M^* M_* g = g.$$

Proof. By Lemma 22 and Proposition 24,

$${}^\perp(\Omega_f^\perp) = {}^\perp \Omega_{M^* f} = \Omega_{M_* M^* f}.$$

Thus Ω_f is a left type if and only if $\Omega_f = \Omega_{M_* M^* f}$, and this is equivalent to $f = M_* M^* f$ by Lemma 22. The right-hand statement is parallel. $\square$

REMARK 27 Scope of the correspondence. The theorem is stated for paired types and their profile coordinates. It is obviously not a bijection between arbitrary subsets of $\mathcal{C} \times \mathbb{R}$ and

arbitrary functions $\mathcal{C} \rightarrow \overline{\mathbb{R}}$; an arbitrary subset is first replaced by the subset Ω_{φ_X} recovered from its profile.

THEOREM 28 Nucleus–types correspondence. *The assignments*

$$(f, g) \longmapsto (\Omega_f, \Omega_g), \quad (A, B) \longmapsto (\lambda_A, \rho_B)$$

are inverse bijections between the objects of $\text{Nuc}(M)$ and the paired types $\text{PType}(M)$. Explicitly,

$$(f, g) \in \text{Nuc}(M) \iff \Omega_f = {}^\perp\Omega_g \text{ and } \Omega_g = \Omega_f^\perp.$$

Proof. Suppose first that $(f, g) \in \text{Nuc}(M)$. Then $g = M^*f$ and $f = M_*g$. Proposition 24 gives

$$\Omega_f^\perp = \Omega_{M^*f} = \Omega_g, \quad {}^\perp\Omega_g = \Omega_{M_*g} = \Omega_f.$$

Hence (Ω_f, Ω_g) is a paired type.

Conversely, let $(A, B) \in \text{PType}(M)$. Thus $B = A^\perp$ and $A = {}^\perp B$. By Proposition 24,

$$B = A^\perp = \Omega_{M^*\lambda_A}, \quad A = {}^\perp B = \Omega_{M_*\rho_B}.$$

Applying profiles and using Lemma 22, together with Corollary 25, gives

$$\rho_B = M^*\lambda_A, \quad \lambda_A = M_*\rho_B.$$

Therefore $(\lambda_A, \rho_B) \in \text{Nuc}(M)$. Finally, $\varphi_{\Omega_f} = f$ for every coordinate f , and paired types are recovered from their profiles by Corollary 25. Thus the two displayed assignments are inverse to one another. $\square$

This completes the binary comparison. Sections 5 through 7 keep the same realisability datum but ask a different question: how the execution product interacts with these closed objects, why the naive products of left and right types do not provide the desired associative calculus, and why the ternary measurement is the appropriate setting for middle types and residual operations.

5. Extending the execution to types

By assumption, the weighted execution product of Section 3 is associative on $\mathcal{C} \times \mathbb{R}$. We now ask how much of this multiplicative structure descends to the binary left and right types generated by weighted orthogonality. The answer is limited: taking a raw product and then closing it produces left and right types, but the resulting closed product need not be associative.

Write

$$(a, \alpha)(b, \beta) = (ab, \alpha + \beta - M(a, b))$$

for the weighted execution product. If $X, Y \subseteq \mathcal{C} \times \mathbb{R}$, define their raw product by

$$XY := \{xy \mid x \in X, y \in Y\}.$$

Since weighted execution is associative, raw products of subsets are associative:

$$(XY)Z = X(YZ).$$

Recall the notation

$$\text{cl}_L(X) := {}^\perp(X^\perp), \quad \text{cl}_R(X) := ({}^\perp X)^\perp$$

for the left and right closures of Section 3. Thus X is a left type precisely when $\text{cl}_L(X) = X$, and a right type precisely when $\text{cl}_R(X) = X$.

DEFINITION 29. For $X, Y \subseteq \mathcal{C} \times \mathbb{R}$, define the *left closed product* and *right closed product* by

$$X \odot_L Y := \text{cl}_L(XY), \quad X \odot_R Y := \text{cl}_R(XY).$$

Thus $X \odot_L Y$ is always a left type and $X \odot_R Y$ is always a right type, by the orthogonal-complement calculus of Section 3. The definition records the closed product used below: first multiply, then apply the appropriate binary closure. Associativity is a separate question.

5.1. The one-sided associativity obstruction

The trefoil identity gives the basic rule for moving a weighted product across orthogonality.

LEMMA 30. For $x = (a, \alpha)$, $y = (b, \beta)$, and $z = (c, \gamma)$ in $\mathcal{C} \times \mathbb{R}$,

$$xy \perp z \iff x \perp yz.$$

Proof. The condition $xy \perp z$ is

$$\alpha + \beta - M(a, b) + \gamma \leq M(ab, c),$$

or equivalently

$$\alpha + \beta + \gamma \leq M(a, b) + M(ab, c).$$

By the trefoil identity (3.1), the right-hand side is $M(b, c) + M(a, bc)$. This is exactly the condition

$$\alpha + \beta + \gamma - M(b, c) \leq M(a, bc),$$

which says $x \perp yz$. □

The next identities isolate the asymmetry that will obstruct associativity.

PROPOSITION 31. For all $X, Y \subseteq \mathcal{C} \times \mathbb{R}$,

$$\text{cl}_L(X) \odot_L Y = X \odot_L Y$$

and

$$X \odot_L Y \subseteq X \odot_L \text{cl}_L(Y).$$

Proof. We prove the first identity by comparing right complements. For $z \in \mathcal{C} \times \mathbb{R}$,

$$\begin{aligned} z \in (\text{cl}_L(X)Y)^\perp &\iff ay \perp z \text{ for all } a \in \text{cl}_L(X), y \in Y \\ &\iff a \perp yz \text{ for all } a \in \text{cl}_L(X), y \in Y \\ &\iff yz \in (\text{cl}_L(X))^\perp \text{ for all } y \in Y. \end{aligned}$$

By triple-complement cancellation, $(\text{cl}_L(X))^\perp = X^\perp$. Therefore the last condition is equivalent to

$$x \perp yz \text{ for all } x \in X, y \in Y,$$

which, by Lemma 30, is equivalent to $xy \perp z$ for all $x \in X$ and $y \in Y$. Hence

$$(\text{cl}_L(X)Y)^\perp = (XY)^\perp.$$

Applying the left complement gives

$$\text{cl}_L(\text{cl}_L(X)Y) = \text{cl}_L(XY),$$

which is the first identity. For the second assertion, $Y \subseteq \text{cl}_L(Y)$, hence $XY \subseteq X \text{cl}_L(Y)$. Taking right complements reverses the inclusion, and taking left complements reverses it again:

$$\text{cl}_L(XY) \subseteq \text{cl}_L(X \text{cl}_L(Y)).$$

This is the desired containment. $\square$

LEMMA 32. For all $X, Y, Z \subseteq \mathcal{C} \times \mathbb{R}$,

$$((X \odot_L Y)Z)^\perp = (X(YZ))^\perp.$$

Proof. Let $t \in \mathcal{C} \times \mathbb{R}$. Then $t \in ((X \odot_L Y)Z)^\perp$ if and only if $dz \perp t$ for every $d \in X \odot_L Y$ and every $z \in Z$. By Lemma 30, this is equivalent to requiring $d \perp zt$ for every such d and z . Since

$$(X \odot_L Y)^\perp = (\text{cl}_L(XY))^\perp = (XY)^\perp$$

by triple-complement cancellation, the preceding condition is equivalent to $zt \in (XY)^\perp$ for every $z \in Z$. Unwinding this, we get $xy \perp zt$ for all $x \in X$, $y \in Y$, and $z \in Z$. Applying Lemma 30 once more, this is equivalent to $x(yz) \perp t$ for all x, y, z , namely $t \in (X(YZ))^\perp$. $\square$

PROPOSITION 33. For all $X, Y, Z \subseteq \mathcal{C} \times \mathbb{R}$,

$$(X \odot_L Y) \odot_L Z = X \odot_L (YZ)$$

and therefore

$$(X \odot_L Y) \odot_L Z \subseteq X \odot_L (Y \odot_L Z).$$

In particular, for a fixed triple X, Y, Z , associativity of the left closed product is equivalent to the equality

$$X \odot_L (YZ) = X \odot_L (Y \odot_L Z).$$

Proof. By Lemma 32,

$$(X \odot_L Y) \odot_L Z = {}^\perp(((X \odot_L Y)Z)^\perp) = {}^\perp((X(YZ))^\perp) = X \odot_L (YZ).$$

Since $YZ \subseteq Y \odot_L Z$, we have $X(YZ) \subseteq X(Y \odot_L Z)$. Applying right and then left complements gives

$$X \odot_L (YZ) \subseteq X \odot_L (Y \odot_L Z).$$

The final assertion follows from the first displayed equality. $\square$

REMARK 34 Scope of the obstruction. This proposition locates the obstruction for the binary left closed product. For a fixed triple, associativity is equivalent to equality in the displayed containment. Questions of functoriality and endpoint behaviour are separate from this statement. The right closed product has a parallel obstruction with the variance reversed.

5.2. A finite non-associativity example

EXAMPLE 35. We now give a finite example in which the containment of Proposition 33 is strict. We'll return to this example for the later unit-profile and two-out-of-three sharpness

calculations. Let

$$\mathcal{C} = \{e, a, c, d\}$$

be the monoid with identity e and multiplication table

$\cdot$	e	a	c	d
e	e	a	c	d
a	a	e	c	d
c	c	d	c	d
d	d	c	c	d

and let

$$p(e) = 0, \quad p(a) = -3, \quad p(c) = 1, \quad p(d) = 4.$$

The associated measurement $M(u, v) = p(uv) - p(u) - p(v)$, with rows and columns ordered as (e, a, c, d) , is

M	e	a	c	d
e	0	0	0	0
a	0	6	3	3
c	0	6	-1	-1
d	0	0	-4	-4

We work in the profile coordinates of Section 4. Thus a vector records a function $\mathcal{C} \rightarrow \overline{\mathbb{R}}$ in the order (e, a, c, d) , and the vector f represents the weighted subset Ω_f . Let

$$S_c = (-\infty, -\infty, 0, -\infty), \quad S_a = (-\infty, 0, -\infty, -\infty).$$

The corresponding subsets are generated, fibrewise downward in the weight coordinate, by $(c, 0)$ and $(a, 0)$. Their left closures are

$$A = M_* M^* S_c = (-6, 0, 0, -6), \quad B = M_* M^* S_a = (-6, 0, -4, -7).$$

For two profiles f, g , the raw product profile in this finite example is

$$\pi_{f,g}(t) := \max_{uv=t} (f(u) + g(v) - M(u, v)),$$

and the left closed product is represented by

$$f \odot_L g := M_* M^* \pi_{f,g}.$$

Here we use the same symbol $\odot_L$ for the operation on profile coordinates and for the corresponding operation on the subsets of the form Ω_f that they represent. Using the displayed matrix for M , a direct calculation gives

$$A \odot_L B = (-6, -3, -3, -6), \quad B \odot_L A = (-6, -3, -3, -6).$$

Continuing the same calculation,

$$(A \odot_L B) \odot_L A = (-8, -2, -2, -8),$$

whereas

$$A \odot_L (B \odot_L A) = (-5, -2, -2, -5).$$

For subsets of the form Ω_f , inclusion is equivalent to pointwise comparison of profiles, so the first profile is strictly smaller than the second. Hence

$$(A \odot_L B) \odot_L A \subsetneq A \odot_L (B \odot_L A),$$

so the left closed product is not associative, even for left types obtained by closing principal weighted elements.

The obstruction exhibited here is the reason the next section passes from binary left and right types to the middle/peripheral framework. There the execution product is tested with both a left and a right context, rather than being forced through one-sided binary closure after each multiplication.

6. Middle types

Section 5 shows that the one-sided closed products of binary left and right types need not be associative, even though the execution on $\mathcal{C} \times \mathbb{R}$ they extend is associative. What is missing from a binary left or right type is information about an element placed between a left context and a right context. This section introduces the corresponding ternary orthogonality relation and its complements.

6.1. The ternary measurement

For $x, b, z \in \mathcal{C}$, define

$$M_3(x, b, z) := p(xbz) - p(x) - p(b) - p(z).$$

This is the measurement of a middle element b placed between a left context x and a right context z . It is related to the binary measurement by the two splittings

$$M_3(x, b, z) = M(x, bz) + M(b, z) = M(xb, z) + M(x, b). \quad (6.1)$$

Both identities follow immediately by expanding the definitions and using associativity of execution. For weighted elements

$$x = (x_0, \xi), \quad b = (b_0, \beta), \quad z = (z_0, \zeta) \quad \text{in } \mathcal{C} \times \mathbb{R},$$

define *middle/peripheral orthogonality* by

$$b \perp\!\!\!\perp (x, z) \iff \xi + \beta + \zeta \leq M_3(x_0, b_0, z_0). \quad (6.2)$$

Thus a single weighted element is tested in the middle position against an ordered pair of peripheral weighted elements.

LEMMA 36. For $x, b, z \in \mathcal{C} \times \mathbb{R}$,

$$b \perp\!\!\!\perp (x, z) \iff xb \perp z \iff x \perp bz.$$

Proof. Write $x = (x_0, \xi)$, $b = (b_0, \beta)$, and $z = (z_0, \zeta)$. The condition $xb \perp z$ is

$$\xi + \beta - M(x_0, b_0) + \zeta \leq M(x_0 b_0, z_0),$$

or equivalently

$$\xi + \beta + \zeta \leq M(x_0, b_0) + M(x_0 b_0, z_0).$$

By (6.1), the right-hand side is $M_3(x_0, b_0, z_0)$. This is exactly $b \perp\!\!\!\perp (x, z)$. The equivalence with $x \perp bz$ is the same calculation using the other splitting in (6.1). $\square$

LEMMA 37. For all $a, a', x, z \in \mathcal{C} \times \mathbb{R}$,

$$aa' \perp\!\!\!\perp (x, z) \iff a \perp\!\!\!\perp (x, a'z) \iff a' \perp\!\!\!\perp (xa, z).$$

Proof. Using Lemma 36 and the execution shift Lemma 30,

$$aa' \perp\!\!\!\perp (x, z) \iff x(aa') \perp z \iff xa \perp a'z \iff a \perp\!\!\!\perp (x, a'z).$$

The same starting condition is also equivalent to

$$x(aa') \perp z \iff (xa)a' \perp z \iff a' \perp\!\!\!\perp (xa, z),$$

where the middle equivalence uses associativity of weighted execution. $\square$

6.2. Middle and peripheral types

We can associate two complement operations to the ternary measurement. For $A \subseteq \mathcal{C} \times \mathbb{R}$, define its peripheral complement by

$$A^\perp := \{(x, z) \in (\mathcal{C} \times \mathbb{R})^2 \mid a \perp\!\!\!\perp (x, z) \text{ for all } a \in A\}.$$

For $P \subseteq (\mathcal{C} \times \mathbb{R})^2$, define its middle complement by

$${}^\perp P := \{a \in \mathcal{C} \times \mathbb{R} \mid a \perp\!\!\!\perp (x, z) \text{ for all } (x, z) \in P\}.$$

DEFINITION 38. For $A \subseteq \mathcal{C} \times \mathbb{R}$ and $P \subseteq (\mathcal{C} \times \mathbb{R})^2$, set

$$\text{cl}_{\text{mid}}(A) := {}^\perp(A^\perp), \quad \text{cl}_{\text{per}}(P) := ({}^\perp P)^\perp.$$

DEFINITION 39. A *middle type* is a subset $A \subseteq \mathcal{C} \times \mathbb{R}$ such that

$$\text{cl}_{\text{mid}}(A) = A.$$

A *peripheral type* is a subset $P \subseteq (\mathcal{C} \times \mathbb{R})^2$ such that

$$\text{cl}_{\text{per}}(P) = P.$$

Equivalently, middle types are the subsets of the form ${}^\perp P$, and peripheral types are the subsets of the form $A^\perp$. The formal complement calculus is the same as in Section 3; we describe it here in the middle-peripheral situation for ease of reference.

PROPOSITION 40 Middle complement calculus. *Let $A, A' \subseteq \mathcal{C} \times \mathbb{R}$ and $P, P' \subseteq (\mathcal{C} \times \mathbb{R})^2$.*

- (i) *If $A \subseteq A'$, then $(A')^\perp \subseteq A^\perp$. If $P \subseteq P'$, then ${}^\perp P' \subseteq {}^\perp P$.*
- (ii) *One has*

$$A \subseteq {}^\perp(A^\perp), \quad P \subseteq ({}^\perp P)^\perp.$$

- (iii) *Alternating three complements cancels to one complement:*

$${}^\perp({}^\perp({}^\perp P)^\perp) = {}^\perp P, \quad ({}^\perp(A^\perp))^\perp = A^\perp.$$

- (iv) *Closure does not change the opposite complement:*

$$(\text{cl}_{\text{mid}} A)^\perp = A^\perp, \quad {}^\perp(\text{cl}_{\text{per}} P) = {}^\perp P.$$

Proof. These are the standard identities for antitone orthogonal-complement operations. Antitonicity follows directly from the definitions: enlarging the set being tested against imposes more inequalities. The double-complement containments follow because every element is orthogonal to all elements in its own complement. Applying antitonicity to these containments gives the reverse inclusions needed for triple-complement cancellation. The final identities are exactly those triple-complement identities applied to the closures in Definition 38. $\square$

6.3. The middle product

If $A, B \subseteq \mathcal{C} \times \mathbb{R}$, let AB denote their raw product under weighted execution:

$$AB := \{ab \mid a \in A, b \in B\}.$$

The *middle closed product* is

$$A \odot B := \text{cl}_{\text{mid}}(AB) = {}^{\perp}((AB)^{\perp}). \quad (6.3)$$

Thus $A \odot B$ is a middle type for arbitrary subsets A and B , and in particular for middle types.

LEMMA 41. For all $A, B, C \subseteq \mathcal{C} \times \mathbb{R}$,

$$((A \odot B)C)^{\perp} = (A(BC))^{\perp}$$

and

$$(A(B \odot C))^{\perp} = (A(BC))^{\perp}.$$

Proof. Let $(x, z) \in (\mathcal{C} \times \mathbb{R})^2$. Then $(x, z) \in ((A \odot B)C)^{\perp}$ if and only if $dc \perp (x, z)$ for every $d \in A \odot B$ and $c \in C$. By Lemma 37, this is equivalent to $d \perp (x, cz)$ for every such d and c . Since

$$(A \odot B)^{\perp} = (\text{cl}_{\text{mid}}(AB))^{\perp} = (AB)^{\perp},$$

the preceding condition is equivalent to requiring $(x, cz) \in (AB)^{\perp}$ for every $c \in C$. Unwinding again, this says $ab \perp (x, cz)$ for all $a \in A$, $b \in B$, and $c \in C$. By Lemma 37, this is equivalent to $a(bc) \perp (x, z)$ for all a, b, c , which is precisely $(x, z) \in (A(BC))^{\perp}$.

The second identity is parallel. The condition $(x, z) \in (A(B \odot C))^{\perp}$ is equivalent to $ae \perp (x, z)$ for every $a \in A$ and $e \in B \odot C$, hence to $e \perp (xa, z)$ for every such a, e . Since $(B \odot C)^{\perp} = (BC)^{\perp}$, this is equivalent to $bc \perp (xa, z)$ for all $b \in B, c \in C$. Applying Lemma 37 once more gives $a(bc) \perp (x, z)$, as required. $\square$

PROPOSITION 42 Associativity of the middle product. For all $A, B, C \subseteq \mathcal{C} \times \mathbb{R}$,

$$(A \odot B) \odot C = A \odot (B \odot C).$$

In particular, the middle closed product restricts to an associative product on middle types.

Proof. By Lemma 41,

$$((A \odot B)C)^{\perp} = (A(BC))^{\perp} = (A(B \odot C))^{\perp}.$$

Applying the middle complement ${}^{\perp}(-)$ to the two outer terms gives

$$(A \odot B) \odot C = {}^{\perp}(((A \odot B)C)^{\perp}) = {}^{\perp}((A(B \odot C))^{\perp}) = A \odot (B \odot C).$$

$\square$

REMARK 43. The same raw product also records edge actions with the binary types of Section 3. If L is a left type and A is a middle type, the left edge action is the Section 5 closed product

$$L \odot_L A := \text{cl}_L(LA).$$

If A is a middle type and R is a right type, the right edge action is

$$A \odot_R R := \text{cl}_R(AR).$$

These actions record how a middle type meets one binary boundary type. Unit or residual laws for them require the additional closure and variance hypotheses stated in the next section.

Section 7 adds the extra structure needed to discuss units and residual operations for the middle product.

7. Units and residual operations

Having constructed the associative middle product from the ternary orthogonality relation, we now record the additional algebraic structure that can be used later in the coordinate calculations: a unit when the execution product has one, and the two oriented residual operations for the middle product.

7.1. Units

The original realisability datum did not include a unit. In this subsection only, assume that the execution product on $\mathcal{C}$ has a two-sided unit ϵ . Set

$$\mathbf{e} := (\epsilon, -p(\epsilon)) \in \mathcal{C} \times \mathbb{R}$$

and define the middle type generated by this weighted element by

$$\mathbf{1} := \text{cl}_{\text{mid}}(\{\mathbf{e}\}) = {}^{\perp}(\{\mathbf{e}\})^{\perp}.$$

LEMMA 44. *The weighted element $\mathbf{e}$ is a two-sided unit for weighted execution on $\mathcal{C} \times \mathbb{R}$.*

Proof. Let $x = (a, \alpha)$. Since $a\epsilon = \epsilon a = a$,

$$\begin{aligned} M(a, \epsilon) &= p(a) - p(a) - p(\epsilon) = -p(\epsilon), \\ M(\epsilon, a) &= p(a) - p(\epsilon) - p(a) = -p(\epsilon). \end{aligned}$$

Hence

$$x\mathbf{e} = (a, \alpha)(\epsilon, -p(\epsilon)) = (a, \alpha),$$

and similarly $\mathbf{e}x = x$. □

PROPOSITION 45. *Under the unit hypothesis above, $\mathbf{1}$ is a two-sided unit for the middle product. For every middle type A ,*

$$\mathbf{1} \odot A = A = A \odot \mathbf{1}.$$

Proof. Since $\{\mathbf{e}\}A = A = A\{\mathbf{e}\}$ by Lemma 44, we have

$$\{\mathbf{e}\} \odot A = \text{cl}_{\text{mid}}(A) = A$$

and similarly $A \odot \{\mathbf{e}\} = A$. Since $\{\mathbf{e}\}\{\mathbf{e}\} = \{\mathbf{e}\}$, we have $\mathbf{1} = \{\mathbf{e}\} \odot \{\mathbf{e}\}$. Proposition 42 then gives

$$\mathbf{1} \odot A = (\{\mathbf{e}\} \odot \{\mathbf{e}\}) \odot A = \{\mathbf{e}\} \odot (\{\mathbf{e}\} \odot A) = A,$$

and the right unit identity is analogous. □

7.2. Left and right units

The same weighted unit also generates one-sided units for the binary closed products of Section 5. Define

$$\mathbf{1}_L := \text{cl}_L(\{\mathbf{e}\}), \quad \mathbf{1}_R := \text{cl}_R(\{\mathbf{e}\}).$$

Thus $\mathbf{1}_L$ is a left type and $\mathbf{1}_R$ is a right type. Their profiles make visible the difference between testing the weighted unit against one boundary and testing it against both boundaries.

PROPOSITION 46 Profiles of the unit types. *Let $f_{\mathbf{1}_L}$, $f_{\mathbf{1}_R}$, and $f_{\mathbf{1}}$ denote the profiles of $\mathbf{1}_L$, $\mathbf{1}_R$, and $\mathbf{1}$, respectively. As $\mathbb{R}$ -valued functions,*

$$\begin{aligned} f_{\mathbf{1}_L}(b) &= \inf_{z \in \mathcal{C}} M(b, z), \\ f_{\mathbf{1}_R}(b) &= \inf_{x \in \mathcal{C}} M(x, b), \\ f_{\mathbf{1}}(b) &= \inf_{x, z \in \mathcal{C}} (M_3(x, b, z) - M(x, z)). \end{aligned}$$

Equivalently,

$$M_3(x, b, z) - M(x, z) = p(xbz) - p(xz) - p(b).$$

Proof. Since $M(\epsilon, z) = -p(\epsilon)$, a weighted element (z, ζ) lies in $\{\mathbf{e}\}^\perp$ if and only if

$$-p(\epsilon) + \zeta \leq -p(\epsilon),$$

equivalently $\zeta \leq 0$. Hence $\{\mathbf{e}\}^\perp$ is the right type with constant profile 0. Therefore $(b, \beta) \in \mathbf{1}_L$ if and only if

$$\beta + \zeta \leq M(b, z) \quad \text{for all } z \in \mathcal{C} \text{ and all } \zeta \leq 0.$$

Since $\zeta = 0$ is allowed, this is equivalent to $\beta \leq \inf_{z \in \mathcal{C}} M(b, z)$. The calculation for $\mathbf{1}_R$ is the same with the two variables interchanged. Namely, ${}^\perp\{\mathbf{e}\}$ has constant profile 0, and taking its right complement gives $f_{\mathbf{1}_R}(b) = \inf_{a \in \mathcal{C}} M(a, b)$.

For the middle unit, first observe that $\{\mathbf{e}\}^\perp$ is exactly the binary orthogonality relation on the two peripheral variables. Indeed, $M_3(x, \epsilon, z) = M(x, z) - p(\epsilon)$, so

$$(\epsilon, -p(\epsilon)) \perp ((x, \xi), (z, \zeta)) \iff \xi + \zeta \leq M(x, z).$$

Hence $(b, \beta) \in \mathbf{1} = {}^\perp(\{\mathbf{e}\}^\perp)$ if and only if

$$\beta + \xi + \zeta \leq M_3(x, b, z)$$

for all $x, z \in \mathcal{C}$ and all $\xi, \zeta \in \mathbb{R}$ with $\xi + \zeta \leq M(x, z)$. Since $\xi + \zeta = M(x, z)$ is allowed, this is equivalent to

$$\beta \leq \inf_{x, z \in \mathcal{C}} (M_3(x, b, z) - M(x, z)).$$

The final displayed identity is obtained by expanding $M_3(x, b, z)$ and $M(x, z)$. □

PROPOSITION 47 Comparison of unit profiles. *Under the unit hypothesis,*

$$f_{\mathbf{1}}(b) \leq f_{\mathbf{1}_L}(b), \quad f_{\mathbf{1}}(b) \leq f_{\mathbf{1}_R}(b)$$

for every $b \in \mathcal{C}$. Equivalently,

$$f_{\mathbf{1}} \leq \min(f_{\mathbf{1}_L}, f_{\mathbf{1}_R})$$

pointwise.

Proof. By Proposition 46,

$$f_1(b) = \inf_{x,z \in \mathcal{C}} (M_3(x, b, z) - M(x, z)).$$

Restricting the infimum to $x = \epsilon$ gives

$$f_1(b) \leq \inf_{z \in \mathcal{C}} (M_3(\epsilon, b, z) - M(\epsilon, z)) = \inf_{z \in \mathcal{C}} M(b, z) = f_{1_L}(b).$$

Restricting the infimum to $z = \epsilon$ gives

$$f_1(b) \leq \inf_{x \in \mathcal{C}} (M_3(x, b, \epsilon) - M(x, \epsilon)) = \inf_{x \in \mathcal{C}} M(x, b) = f_{1_R}(b).$$

□

PROPOSITION 48 One-sided unit laws. *If A is a left type, then*

$$\mathbf{1}_L \odot_L A = A.$$

If B is a right type, then

$$B \odot_R \mathbf{1}_R = B.$$

Proof. The left identity follows from Proposition 31 and Lemma 44:

$$\mathbf{1}_L \odot_L A = \text{cl}_L(\{\mathbf{e}\}) \odot_L A = \{\mathbf{e}\} \odot_L A = \text{cl}_L(A) = A.$$

For the right identity we use the right-hand analogue

$$X \odot_R \text{cl}_R(Y) = X \odot_R Y.$$

Indeed, comparing left complements, $q \in {}^\perp(X \text{cl}_R(Y))$ if and only if $q \perp xy$ for all $x \in X$ and $y \in \text{cl}_R(Y)$. By Lemma 30, this is equivalent to $qx \perp y$ for all such x and y , or $qx \in {}^\perp \text{cl}_R(Y) = {}^\perp Y$ for all $x \in X$. This is equivalent, again by Lemma 30, to $q \in {}^\perp(XY)$. Taking right complements gives the displayed identity. Hence, for a right type B ,

$$B \odot_R \mathbf{1}_R = B \odot_R \text{cl}_R(\{\mathbf{e}\}) = B \odot_R \{\mathbf{e}\} = \text{cl}_R(B) = B.$$

□

REMARK 49. The preceding proposition is deliberately one-sided. It gives a left unit for the left closed product and a right unit for the right closed product. The identities $A \odot_L \mathbf{1}_L = A$ and $\mathbf{1}_R \odot_R B = B$ would require control of closure in the second variable of $\odot_L$, respectively the first variable of $\odot_R$. Section 5 isolates this one-sided closure issue.

EXAMPLE 50 The unit profiles in the running example. Return to the finite monoid of Example 35. Its unit is e , and since $p(e) = 0$, the weighted unit is $\mathbf{e} = (e, 0)$. From the displayed matrix for M , the row minima and column minima give the one-sided unit profiles, and the middle unit formula gives the middle profile. In the order (e, a, c, d) ,

$$f_{1_L} = (0, 0, -1, -4), \quad f_{1_R} = (0, 0, -4, -4), \quad f_1 = (0, 0, -4, -7).$$

Thus the running example distinguishes the left, right, and middle unit profiles. It also shows that the comparison in Proposition 47 can be strict: at d ,

$$f_1(d) = -7 < -4 = \min(f_{1_L}(d), f_{1_R}(d)).$$

The value -7 is attained in the middle unit formula by the peripheral pair $(x, z) = (c, a)$: the multiplication changes from $ca = d$ to $cda = c$.

7.3. Residuals

The middle product is not commutative, so it has two oriented residuals. We first define them as subsets of $\mathcal{C} \times \mathbb{R}$.

DEFINITION 51. Let $A \subseteq \mathcal{C} \times \mathbb{R}$ and let B be a middle type. Define

$$A \multimap_l B := \{f \in \mathcal{C} \times \mathbb{R} \mid af \in B \text{ for all } a \in A\},$$

and

$$A \multimap_r B := \{f \in \mathcal{C} \times \mathbb{R} \mid fa \in B \text{ for all } a \in A\}.$$

For residuals of middle types we also need two actions of a middle subset on a peripheral subset. If $A \subseteq \mathcal{C} \times \mathbb{R}$ and $P \subseteq (\mathcal{C} \times \mathbb{R})^2$, set

$$A \odot_l P := \{(xa, z) \mid a \in A, (x, z) \in P\},$$

and

$$A \odot_r P := \{(x, az) \mid a \in A, (x, z) \in P\}.$$

These are raw peripheral subsets; no peripheral closure is included in the notation.

PROPOSITION 52 Residuals as middle types. *If $A \subseteq \mathcal{C} \times \mathbb{R}$ and B is a middle type, then*

$$A \multimap_l B = {}^\perp(A \odot_l B^\perp), \quad A \multimap_r B = {}^\perp(A \odot_r B^\perp).$$

In particular, $A \multimap_l B$ and $A \multimap_r B$ are middle types.

Proof. We prove the left residual formula. For $f \in \mathcal{C} \times \mathbb{R}$,

$$f \in A \multimap_l B$$

means that $af \in B$ for every $a \in A$. Since B is a middle type, this is equivalent to

$$af \perp (x, z) \quad \text{for all } a \in A \text{ and } (x, z) \in B^\perp.$$

By Lemma 37, the displayed condition is equivalent to

$$f \perp (xa, z) \quad \text{for all } a \in A \text{ and } (x, z) \in B^\perp,$$

which is precisely $f \in {}^\perp(A \odot_l B^\perp)$.

The proof of the right residual formula is the same: $fa \perp (x, z)$ is equivalent to $f \perp (x, az)$ by Lemma 37. $\square$

PROPOSITION 53 Residuation. *Let A, B, C be middle types. Then*

$$A \odot B \subseteq C \iff B \subseteq A \multimap_l C \iff A \subseteq B \multimap_r C.$$

Proof. Since C is middle closed,

$$A \odot B = \text{cl}_{\text{mid}}(AB) \subseteq C \iff AB \subseteq C.$$

The latter condition says that $ab \in C$ for all $a \in A$ and $b \in B$. Equivalently, every $b \in B$ lies in $A \multimap_l C$, and equivalently every $a \in A$ lies in $B \multimap_r C$. $\square$

PROPOSITION 54 Partial transitivity. *Let A, B, C be middle types. Then*

$$(A \multimap_l B) \odot (B \multimap_l C) \subseteq A \multimap_l C$$

and

$$(B \multimap_r C) \odot (A \multimap_r B) \subseteq A \multimap_r C.$$

Proof. We prove the left-handed statement. Since $A \multimap_l C$ is a middle type by Proposition 52, it is enough to show that the raw product

$$(A \multimap_l B)(B \multimap_l C)$$

is contained in $A \multimap_l C$. Let $g \in A \multimap_l B$ and $h \in B \multimap_l C$. For each $a \in A$, one has $ag \in B$, and therefore $(ag)h \in C$. Associativity of weighted execution gives $a(gh) = (ag)h$, so $gh \in A \multimap_l C$. The right-handed statement is parallel. If $g \in B \multimap_r C$ and $h \in A \multimap_r B$, then for $a \in A$ one has $ha \in B$, and hence $g(ha) \in C$. Associativity gives $(gh)a = g(ha)$, so $gh \in A \multimap_r C$. $\square$

LEMMA 55. *Assume the execution product has a unit, and let $\mathbf{e}$ be the weighted unit of Lemma 44. If A and B are middle types, then*

$$A \subseteq B \iff \mathbf{e} \in A \multimap_l B \iff \mathbf{e} \in A \multimap_r B.$$

Proof. This is immediate from $a\mathbf{e} = a = \mathbf{e}a$ for every $a \in \mathcal{C} \times \mathbb{R}$. $\square$

7.4. The assembled type calculus

The constructions of Sections 5–7 are organized by the closure applied after raw weighted execution. Let

$$\mathbf{L} = \text{Fix}(\text{cl}_L), \quad \mathbf{M} = \text{Fix}(\text{cl}_{\text{mid}}), \quad \mathbf{R} = \text{Fix}(\text{cl}_R)$$

denote the left, middle, and right types. For arbitrary $X, Y \subseteq \mathcal{C} \times \mathbb{R}$, the same raw product XY has three closed forms:

$$X \odot_L Y = \text{cl}_L(XY) \in \mathbf{L}, \quad X \odot Y = \text{cl}_{\text{mid}}(XY) \in \mathbf{M}, \quad X \odot_R Y = \text{cl}_R(XY) \in \mathbf{R}.$$

The product symbol keeps track of which closure is used. The middle types carry the associative core of the calculus. Ordered by inclusion, middle types carry the following Lambek-style structure.

PROPOSITION 56 The assembled middle calculus. *Let A, B, C be middle types.*

- (i) *The product $\odot$ is monotone in both variables and associative. If the execution product has a unit, then*

$$\mathbf{1} \odot A = A = A \odot \mathbf{1}.$$

- (ii) *The residuals are middle types and are characterized by*

$$A \odot B \subseteq C \iff B \subseteq A \multimap_l C \iff A \subseteq B \multimap_r C.$$

- (iii) *The residuals curry the product:*

$$(A \odot B) \multimap_l C = B \multimap_l (A \multimap_l C),$$

and

$$(A \odot B) \multimap_r C = A \multimap_r (B \multimap_r C).$$

(iv) *The two orientations satisfy the mixed identity*

$$A \multimap_l (B \multimap_r C) = B \multimap_r (A \multimap_l C).$$

(v) *Residual arrows compose by the inclusions*

$$(A \multimap_l B) \odot (B \multimap_l C) \subseteq A \multimap_l C,$$

and

$$(B \multimap_r C) \odot (A \multimap_r B) \subseteq A \multimap_r C.$$

(vi) *Under the unit hypothesis, inclusion is detected by the middle unit:*

$$A \subseteq B \iff \mathbf{1} \subseteq A \multimap_l B \iff \mathbf{1} \subseteq A \multimap_r B.$$

Proof. Monotonicity follows from monotonicity of raw product and of middle closure. Associativity is Proposition 42, the unit law is Proposition 45, the fact that residuals are middle types is Proposition 52, and the residual characterization is Proposition 53.

The currying identities and the mixed identity are formal consequences of associativity and residuation. Let D be a middle type. Then

$$\begin{aligned} D \subseteq (A \odot B) \multimap_l C &\iff (A \odot B) \odot D \subseteq C \\ &\iff A \odot (B \odot D) \subseteq C \\ &\iff B \odot D \subseteq A \multimap_l C \\ &\iff D \subseteq B \multimap_l (A \multimap_l C). \end{aligned}$$

Similarly,

$$\begin{aligned} D \subseteq (A \odot B) \multimap_r C &\iff D \odot (A \odot B) \subseteq C \\ &\iff (D \odot A) \odot B \subseteq C \\ &\iff D \odot A \subseteq B \multimap_r C \\ &\iff D \subseteq A \multimap_r (B \multimap_r C), \end{aligned}$$

and

$$\begin{aligned} D \subseteq A \multimap_l (B \multimap_r C) &\iff A \odot D \subseteq B \multimap_r C \\ &\iff (A \odot D) \odot B \subseteq C \\ &\iff A \odot (D \odot B) \subseteq C \\ &\iff D \odot B \subseteq A \multimap_l C \\ &\iff D \subseteq B \multimap_r (A \multimap_l C). \end{aligned}$$

Since the two sides compared in each displayed identity are middle types, testing against all middle D gives equality. The composition inclusions are Proposition 54. Finally, Lemma 55 detects inclusion by membership of the weighted unit $\mathbf{e}$ in either residual; because the residuals are middle types, this is equivalent to containing $\text{cl}_{\text{mid}}(\{\mathbf{e}\}) = \mathbf{1}$. $\square$

The boundary products sit alongside this middle calculus as one-sided boundary forms of the same execution product. If $X_L \in \mathbf{L}$, $A \in \mathbf{M}$, and $X_R \in \mathbf{R}$, then the raw products can be closed in the middle sort,

$$X_L \odot A = \text{cl}_{\text{mid}}(X_L A), \quad A \odot X_R = \text{cl}_{\text{mid}}(A X_R), \quad X_L \odot X_R = \text{cl}_{\text{mid}}(X_L X_R),$$

or at the boundary,

$$X_L \odot_L A = \text{cl}_L(X_L A), \quad A \odot_R X_R = \text{cl}_R(A X_R).$$

Under the unit hypothesis, the boundary unit laws proved above are

$$\mathbf{1}_L \odot_L X_L = X_L, \quad X_R \odot_R \mathbf{1}_R = X_R.$$

Further boundary identities are governed by the containment

$$X \odot_L (YZ) \subseteq X \odot_L (Y \odot_L Z)$$

from Proposition 33, together with the right-handed analogue. Closing an intermediate product at a boundary can change the data seen by the next boundary product.

The output of the section is therefore a three-level type calculus, with the middle level as its associative and residuated part. Section 8 changes representation. There the ternary measurement is regarded as the profunctor

$$N: \mathcal{C} \nrightarrow \mathcal{C} \times \mathcal{C}, \quad N(b, (x, z)) = M_3(x, b, z),$$

and a middle type A is written in profile coordinates. If $f = \varphi_A$, then the boundary profile determined by $A^\perp$ is

$$g = N^*f: \mathcal{C} \times \mathcal{C} \rightarrow \overline{\mathbb{R}}.$$

Thus the same middle type paired with its periphery type is represented by the nuclear point

$$(f, g) = (f, N^*f) \in \text{Nuc}(N).$$

The second coordinate is a binary kernel in the two boundary variables. Viewed as a profunctor $\mathcal{C} \nrightarrow \mathcal{C}$, it has its own Isbell nucleus. Section 8 studies these derived nuclei and rewrites the product and residuals assembled above in those coordinates.

The preceding propositions exhibit the middle part of a larger asymmetric calculus involving left, right, and middle types. The operation $\odot$ is used here as an order-theoretic product on middle types: its functorial content is expressed by the inclusion and residuation laws above. The left and right type operations record the one-sided boundary behaviour of execution, while middle types record execution with both boundaries present. This asymmetric calculus is the structure transported to presheaf coordinates in Section 8.

REMARK 57. Further operations can be considered between peripheral types, but their closure and variance hypotheses require a separate analysis. The present section records the unit and residual structure needed for the middle calculus.

8. Derived nuclei in presheaf coordinates

The ternary measurement defines an $\overline{\mathbb{R}}$ -profunctor

$$N: \mathcal{C} \nrightarrow \mathcal{C} \times \mathcal{C}, \quad N(b, (x, z)) = M_3(x, b, z),$$

where

$$M_3(x, b, z) = p(xbz) - p(x) - p(b) - p(z).$$

We have two pictures of middle types. In the type picture, a middle type is a subset of $\mathcal{C} \times \mathbb{R}$ closed under $\perp$ -biorthogonality; Sections 6–7 built the Lambek calculus of middle types in that form. In the Isbell picture, the same middle type is represented, using the profile dictionary of Section 4, by a point

$$(f, g) \in \text{Nuc}(N),$$

where $f: \mathcal{C} \rightarrow \overline{\mathbb{R}}$ is the presheaf coordinate and $g = N^*f: \mathcal{C} \times \mathcal{C} \rightarrow \overline{\mathbb{R}}$ is the copresheaf coordinate. The second coordinate g is itself a binary profunctor

$$g: \mathcal{C} \nrightarrow \mathcal{C},$$

which has its own Isbell nucleus. We call $\text{Nuc}(g)$ the nucleus *derived* from the middle nuclear point (f, g) . If this point comes from a middle type A , so that $f = \varphi_A$ and $g = N^*f$, we also call $\text{Nuc}(g)$ the nucleus derived from A . Its points will be called derived types.

The first result of the section shows that the nucleus derived from a middle point is computed by the two residuals $f \multimap_l$ and $f \multimap_r$ of the Lambek calculus. The ternary measurement M_3 also has two other one-variable arrangements, obtained by singling out the left or the right coordinate rather than the middle one. These give two further iterated nuclei. The three constructions are then compared by a two-out-of-three theorem: compatibility with any two arrangements forces compatibility with the third. Their common intersection is the balanced locus computed below.

We begin by recording the formulas that translate the type calculus into Isbell coordinates. It will be convenient to recall the trefoil identities:

$$M_3(x, b, z) = M(x, bz) + M(b, z) = M(xb, z) + M(x, b). \quad (8.1)$$

There are higher splitting identities as well. For

$$M_4(a, b, c, d) := p(abcd) - p(a) - p(b) - p(c) - p(d),$$

two adjacent variables may be contracted to their product:

$$\begin{aligned} M_4(a, b, c, d) &= M_3(ab, c, d) + M(a, b) \\ &= M_3(a, bc, d) + M(b, c) \\ &= M_3(a, b, cd) + M(c, d). \end{aligned} \quad (8.2)$$

Both (8.1) and (8.2) follow immediately from the definitions of the measurements and associativity of the execution product.

We shall use one raw convolution operation on one-variable functions. For $r, s: \mathcal{C} \rightarrow \overline{\mathbb{R}}$, define

$$(r \star s)(y) := \sup_{ab=y} (r(a) + s(b) - M(a, b)), \quad (8.3)$$

with $\sup \emptyset = -\infty$. This operation is not itself a type constructor; it is a coordinate-level operation. Type constructors are obtained by applying the relevant Isbell closures.

For example, let A and A' be middle types, and write

$$f = \varphi_A, \quad f' = \varphi_{A'}$$

for their presheaf coordinates. Let

$$A'' = A \odot A', \quad f'' = \varphi_{A''}.$$

The product $A'' = A \odot A'$ is obtained by closing the raw coordinate $f \star f'$ in the middle nucleus:

$$f'' = N_* N^*(f \star f').$$

The copresheaf coordinate of this product has a single M_4 -formula. Define

$$P_{f, f'}(x, z) := \inf_{a, a' \in \mathcal{C}} (M_4(x, a, a', z) - f(a) - f'(a')). \quad (8.4)$$

Then

$$N^*(f \star f') = P_{f, f'}. \quad (8.5)$$

Indeed,

$$\begin{aligned} N^*(f \star f')(x, z) &= \inf_y (M_3(x, y, z) - (f \star f')(y)) \\ &= \inf_{a, a'} (M_3(x, aa', z) + M(a, a') - f(a) - f'(a')) \\ &= \inf_{a, a'} (M_4(x, a, a', z) - f(a) - f'(a')) \\ &= P_{f, f'}(x, z). \end{aligned}$$

Here we use $r - \sup_i s_i = \inf_i (r - s_i)$ and the middle contraction in (8.2). Consequently

$$f'' = N_* P_{f,f'}, \quad A'' = \Omega_{N_* P_{f,f'}}.$$

Since $P_{f,f'} = N^*(f \star f')$, the pair

$$(N_* P_{f,f'}, P_{f,f'})$$

lies in $\text{Nuc}(N)$. This is the middle nuclear point corresponding to $A \odot A'$. Its copresheaf coordinate $\mathcal{C} \times \mathcal{C} \rightarrow \overline{\mathbb{R}}$ is the kernel $P_{f,f'}$.

Finally, the residuals of the raw convolution are given by

$$(f \multimap_l r)(z) = \inf_{b \in \mathcal{C}} (r(bz) + M(b, z) - f(b)),$$

and

$$(f \multimap_r \ell)(x) = \inf_{b \in \mathcal{C}} (\ell(xb) + M(x, b) - f(b)).$$

They are characterized by the residuation laws

$$f \star v \leq r \iff v \leq f \multimap_l r,$$

and

$$u \star f \leq \ell \iff u \leq f \multimap_r \ell.$$

When the inputs are coordinates of the appropriate types, these coordinate operations are the two residuals of the middle Lambek calculus.

8.1. The middle derived nucleus

The first calculation concerns the derived nucleus of a middle nuclear point.

THEOREM 58 Derived residual formulas. *Let $(f, g) \in \text{Nuc}(N)$, and let $(u, v) \in \text{Nuc}(g)$. Then*

$$v = f \multimap_l M^* u, \quad u = f \multimap_r M_* v.$$

Proof. Since $(f, g) \in \text{Nuc}(N)$, we have

$$g(x, z) = \inf_b (M_3(x, b, z) - f(b)).$$

Since $(u, v) \in \text{Nuc}(g)$, we have

$$v(z) = \inf_x (g(x, z) - u(x)), \quad u(x) = \inf_z (g(x, z) - v(z)).$$

Therefore

$$v(z) = \inf_{x,b} (M_3(x, b, z) - u(x) - f(b)).$$

Using the trefoil identity

$$M_3(x, b, z) = M(x, bz) + M(b, z),$$

we obtain

$$\begin{aligned} v(z) &= \inf_{x,b} (M(x, bz) + M(b, z) - u(x) - f(b)) \\ &= \inf_b ((M^* u)(bz) + M(b, z) - f(b)) \\ &= (f \multimap_l M^* u)(z). \end{aligned}$$

Similarly,

$$u(x) = \inf_{b,z} (M_3(x, b, z) - f(b) - v(z)).$$

Using the other trefoil identity

$$M_3(x, b, z) = M(xb, z) + M(x, b),$$

gives

$$\begin{aligned} u(x) &= \inf_{b,z} (M(xb, z) + M(x, b) - f(b) - v(z)) \\ &= \inf_b ((M_*v)(xb) + M(x, b) - f(b)) \\ &= (f \circ_r M_*v)(x). \end{aligned}$$

□

COROLLARY 59 Binary nuclearity of derived coordinates. *Let $(f, g) \in \text{Nuc}(N)$, and let $(u, v) \in \text{Nuc}(g)$. Then*

$$u = M_*(f \star v), \quad v = M^*(u \star f).$$

Consequently

$$M_*M^*u = u, \quad M^*M_*v = v.$$

Thus the two coordinates of a derived type are separately fixed by the binary Isbell closures:

$$u \in \text{Fix}(M_*M^*), \quad v \in \text{Fix}(M^*M_*).$$

Equivalently,

$$(u, M^*u) \in \text{Nuc}(M), \quad (M_*v, v) \in \text{Nuc}(M).$$

The derived pair (u, v) need not itself lie in $\text{Nuc}(M)$.

Proof. From the proof of Theorem 58,

$$u(x) = \inf_{b,z} (M_3(x, b, z) - f(b) - v(z)).$$

Using

$$M_3(x, b, z) = M(x, bz) + M(b, z),$$

and regrouping by $y = bz$, we obtain

$$\begin{aligned} u(x) &= \inf_{b,z} (M(x, bz) + M(b, z) - f(b) - v(z)) \\ &= \inf_y \left(M(x, y) - \sup_{bz=y} (f(b) + v(z) - M(b, z)) \right) \\ &= M_*(f \star v)(x). \end{aligned}$$

Similarly, from the proof of Theorem 58,

$$v(z) = \inf_{x,b} (M_3(x, b, z) - u(x) - f(b)).$$

Using

$$M_3(x, b, z) = M(xb, z) + M(x, b),$$

and regrouping by $y = xb$, we obtain

$$\begin{aligned} v(z) &= \inf_{x,b} (M(xb, z) + M(x, b) - u(x) - f(b)) \\ &= \inf_y \left(M(y, z) - \sup_{xb=y} (u(x) + f(b) - M(x, b)) \right) \\ &= M^*(u \star f)(z). \end{aligned}$$

Thus u lies in the image of M_* , and v lies in the image of M^* . By the fixed-point characterization of the Isbell images,

$$M_*M^*u = u, \quad M^*M_*v = v.$$

□

Thus the derived nucleus of a middle point is computed by the two residuals of the middle Lambek calculus. When the point (u, v) is a type derived from a middle type A , the types Ω_u and Ω_v are left and right types of the original realisability situation $(\mathcal{C}, \text{Ex}, p)$.

8.2. The two boundary arrangements

The ternary measurement M_3 has two other one-variable arrangements besides the middle-periphery arrangement N :

$$L(x, (b, z)) := M_3(x, b, z), \quad R(z, (x, b)) := M_3(x, b, z).$$

THEOREM 60 Boundary residual formulas. *The following formulas hold.*
(L) *Let $(u, h) \in \text{Nuc}(L)$, and let $(f, v) \in \text{Nuc}(h)$. Then*

$$v = f \multimap_l M^*u.$$

Equivalently,

$$v(z) = \inf_b ((M^*u)(bz) + M(b, z) - f(b)).$$

The middle coordinate also satisfies the two boundary formulas

$$f(b) = \inf_x ((M_*v)(xb) + M(x, b) - u(x)) = \inf_z ((M^*u)(bz) + M(b, z) - v(z)).$$

(R) *Let $(v, k) \in \text{Nuc}(R)$, and let $(u, f) \in \text{Nuc}(k)$. Then*

$$u = f \multimap_r M_*v.$$

Equivalently,

$$u(x) = \inf_b ((M_*v)(xb) + M(x, b) - f(b)).$$

The middle coordinate satisfies the same two boundary formulas

$$f(b) = \inf_x ((M_*v)(xb) + M(x, b) - u(x)) = \inf_z ((M^*u)(bz) + M(b, z) - v(z)).$$

Proof. For the left arrangement, since $(u, h) \in \text{Nuc}(L)$,

$$h(b, z) = \inf_x (M_3(x, b, z) - u(x)).$$

Since $(f, v) \in \text{Nuc}(h)$,

$$f(b) = \inf_z (h(b, z) - v(z)), \quad v(z) = \inf_b (h(b, z) - f(b)).$$

Substitution gives

$$f(b) = \inf_{x, z} (M_3(x, b, z) - u(x) - v(z)),$$

and

$$v(z) = \inf_{x, b} (M_3(x, b, z) - u(x) - f(b)).$$

Using

$$M_3(x, b, z) = M(x, bz) + M(b, z),$$

in the formula for v , we obtain

$$\begin{aligned} v(z) &= \inf_{x,b} (M(x, bz) + M(b, z) - u(x) - f(b)) \\ &= \inf_b ((M^*u)(bz) + M(b, z) - f(b)) \\ &= (f \multimap_l M^*u)(z). \end{aligned}$$

The two formulas for f follow from the two trefoil identities. Using

$$M_3(x, b, z) = M(xb, z) + M(x, b),$$

gives

$$f(b) = \inf_x ((M_*v)(xb) + M(x, b) - u(x)),$$

while using

$$M_3(x, b, z) = M(x, bz) + M(b, z)$$

gives

$$f(b) = \inf_z ((M^*u)(bz) + M(b, z) - v(z)).$$

The right arrangement is analogous. Since $(v, k) \in \text{Nuc}(R)$,

$$k(x, b) = \inf_z (M_3(x, b, z) - v(z)).$$

Since $(u, f) \in \text{Nuc}(k)$,

$$u(x) = \inf_b (k(x, b) - f(b)), \quad f(b) = \inf_x (k(x, b) - u(x)).$$

Hence

$$u(x) = \inf_{b,z} (M_3(x, b, z) - f(b) - v(z)),$$

and

$$f(b) = \inf_{x,z} (M_3(x, b, z) - u(x) - v(z)).$$

Using

$$M_3(x, b, z) = M(xb, z) + M(x, b)$$

in the formula for u , we get

$$\begin{aligned} u(x) &= \inf_{b,z} (M(xb, z) + M(x, b) - f(b) - v(z)) \\ &= \inf_b ((M_*v)(xb) + M(x, b) - f(b)) \\ &= (f \multimap_r M_*v)(x). \end{aligned}$$

The two formulas for f are obtained from the same two trefoil identities, exactly as in the left arrangement. $\square$

8.3. Two-out-of-three coherence

Thus the three one-variable arrangements of M_3 give three ways of producing triples (u, f, v) . The middle arrangement starts from a middle coordinate f and produces the two boundary coordinates. The left boundary arrangement starts from u and produces f and v . The right boundary arrangement starts from v and produces u and f . We now compare the resulting triples. Let

$$\begin{aligned} T_L &:= \{(u, f, v) \mid (u, L^*u) \in \text{Nuc}(L), (f, v) \in \text{Nuc}(L^*u)\}, \\ T_N &:= \{(u, f, v) \mid (f, N^*f) \in \text{Nuc}(N), (u, v) \in \text{Nuc}(N^*f)\}, \\ T_R &:= \{(u, f, v) \mid (v, R^*v) \in \text{Nuc}(R), (u, f) \in \text{Nuc}(R^*v)\}. \end{aligned}$$

These are the triples obtained by iterating the nuclei of the left boundary, middle, and right boundary arrangements. For a triple (u, f, v) , consider the following three residual equations:

$$u(x) = \inf_{b,z} (M_3(x, b, z) - f(b) - v(z)), \quad (B_u)$$

$$f(b) = \inf_{x,z} (M_3(x, b, z) - u(x) - v(z)), \quad (B_f)$$

$$v(z) = \inf_{x,b} (M_3(x, b, z) - u(x) - f(b)). \quad (B_v)$$

We call (u, f, v) a balanced triple for M_3 if it satisfies all three equations. Let T_{bal} denote the set of balanced triples.

LEMMA 61 Equations seen by the three arrangements. *The three arrangements satisfy the following implications:*

$$(u, f, v) \in T_L \implies (B_f) \text{ and } (B_v),$$

$$(u, f, v) \in T_N \implies (B_u) \text{ and } (B_v),$$

$$(u, f, v) \in T_R \implies (B_u) \text{ and } (B_f).$$

Proof. Suppose first that $(u, f, v) \in T_L$. Put $h = L^*u$. Then

$$h(b, z) = \inf_x (M_3(x, b, z) - u(x)),$$

and $(f, v) \in \text{Nuc}(h)$. Hence

$$\begin{aligned} f(b) &= \inf_z (h(b, z) - v(z)) \\ &= \inf_{x,z} (M_3(x, b, z) - u(x) - v(z)), \end{aligned}$$

and

$$\begin{aligned} v(z) &= \inf_b (h(b, z) - f(b)) \\ &= \inf_{x,b} (M_3(x, b, z) - u(x) - f(b)). \end{aligned}$$

Thus (B_f) and (B_v) hold. The middle arrangement is the same calculation with $g = N^*f$. If $(u, f, v) \in T_N$, then

$$g(x, z) = \inf_b (M_3(x, b, z) - f(b)),$$

and $(u, v) \in \text{Nuc}(g)$. Therefore

$$u(x) = \inf_{b,z} (M_3(x, b, z) - f(b) - v(z)),$$

and

$$v(z) = \inf_{x,b} (M_3(x, b, z) - u(x) - f(b)).$$

Thus (B_u) and (B_v) hold. Finally, suppose that $(u, f, v) \in T_R$. Put $k = R^*v$. Then

$$k(x, b) = \inf_z (M_3(x, b, z) - v(z)),$$

and $(u, f) \in \text{Nuc}(k)$. Hence

$$u(x) = \inf_{b,z} (M_3(x, b, z) - f(b) - v(z)),$$

and

$$f(b) = \inf_{x,z} (M_3(x, b, z) - u(x) - v(z)).$$

Thus (B_u) and (B_f) hold. □

THEOREM 62 Two-out-of-three coherence. *The three iterated ternary constructions have the same pairwise intersection:*

$$T_L \cap T_N = T_N \cap T_R = T_L \cap T_R = T_L \cap T_N \cap T_R = T_{\text{bal}}.$$

Proof. By Lemma 61, membership in any two of T_L, T_N, T_R gives all three equations (B_u) , (B_f) , and (B_v) . Hence every pairwise intersection is contained in T_{bal} . Conversely, suppose that $(u, f, v) \in T_{\text{bal}}$. We show that it lies in all three arrangement sets. First put

$$g = N^*f.$$

Then

$$g(x, z) = \inf_b (M_3(x, b, z) - f(b)).$$

Equations (B_u) and (B_v) say exactly that

$$u(x) = \inf_z (g(x, z) - v(z)), \quad v(z) = \inf_x (g(x, z) - u(x)).$$

Thus $(u, v) \in \text{Nuc}(g)$, once we know that $(f, g) \in \text{Nuc}(N)$. Since $g = N^*f$, the closure inequality gives

$$f \leq N_*g.$$

Equation (B_v) implies that, for all x, b, z ,

$$u(x) + v(z) \leq M_3(x, b, z) - f(b).$$

Taking the infimum over b gives

$$u(x) + v(z) \leq g(x, z).$$

Therefore

$$\begin{aligned} N_*g(b) &= \inf_{x,z} (M_3(x, b, z) - g(x, z)) \\ &\leq \inf_{x,z} (M_3(x, b, z) - u(x) - v(z)) \\ &= f(b), \end{aligned}$$

where the last equality is (B_f) . Hence

$$f = N_*g.$$

Thus $(f, g) \in \text{Nuc}(N)$, and consequently

$$(u, f, v) \in T_N.$$

Next put

$$h = L^*u.$$

Then

$$h(b, z) = \inf_x (M_3(x, b, z) - u(x)).$$

Equations (B_f) and (B_v) say exactly that

$$f(b) = \inf_z (h(b, z) - v(z)), \quad v(z) = \inf_b (h(b, z) - f(b)).$$

Thus $(f, v) \in \text{Nuc}(h)$, once we know that $(u, h) \in \text{Nuc}(L)$. Since $h = L^*u$, the closure inequality gives

$$u \leq L_*h.$$

Equation (B_f) implies that, for all x, b, z ,

$$f(b) + v(z) \leq M_3(x, b, z) - u(x).$$

Taking the infimum over x gives

$$f(b) + v(z) \leq h(b, z).$$

Therefore

$$\begin{aligned} L_*h(x) &= \inf_{b,z} (M_3(x, b, z) - h(b, z)) \\ &\leq \inf_{b,z} (M_3(x, b, z) - f(b) - v(z)) \\ &= u(x), \end{aligned}$$

where the last equality is (B_u) . Hence

$$u = L_*h.$$

Thus $(u, h) \in \text{Nuc}(L)$, and consequently

$$(u, f, v) \in T_L.$$

Finally put

$$k = R^*v.$$

Then

$$k(x, b) = \inf_z (M_3(x, b, z) - v(z)).$$

Equations (B_u) and (B_f) say exactly that

$$u(x) = \inf_b (k(x, b) - f(b)), \quad f(b) = \inf_x (k(x, b) - u(x)).$$

Thus $(u, f) \in \text{Nuc}(k)$, once we know that $(v, k) \in \text{Nuc}(R)$. Since $k = R^*v$, the closure inequality gives

$$v \leq R_*k.$$

Equation (B_u) implies that, for all x, b, z ,

$$u(x) + f(b) \leq M_3(x, b, z) - v(z).$$

Taking the infimum over z gives

$$u(x) + f(b) \leq k(x, b).$$

Therefore

$$\begin{aligned} R_*k(z) &= \inf_{x,b} (M_3(x, b, z) - k(x, b)) \\ &\leq \inf_{x,b} (M_3(x, b, z) - u(x) - f(b)) \\ &= v(z), \end{aligned}$$

where the last equality is (B_v) . Hence

$$v = R_*k.$$

Thus $(v, k) \in \text{Nuc}(R)$, and consequently

$$(u, f, v) \in T_R.$$

Therefore every balanced triple lies in $T_L \cap T_N \cap T_R$. The claimed equalities follow. $\square$

The common intersection is therefore explicit: it is the locus where the three one-coordinate residual equations of M_3 are simultaneously saturated. Equivalently, on T_{bal} , each coordinate is recovered from the other two:

$$u(x) = \inf_{b,z} (M_3(x, b, z) - f(b) - v(z)),$$

$$f(b) = \inf_{x,z} (M_3(x, b, z) - u(x) - v(z)),$$

and

$$v(z) = \inf_{x,b} (M_3(x, b, z) - u(x) - f(b)).$$

Thus any one coordinate may be taken as the coordinate from which the other two are derived: (f, v) is derived from u in the left arrangement, (u, v) is derived from f in the middle arrangement, and (u, f) is derived from v in the right arrangement. Moreover, within the balanced locus, the derived pair uniquely recovers the coordinate from which it is derived.

The next example shows why the last qualification is necessary. A pair of coordinates can be derived from a middle coordinate f , while the middle coordinate recovered from that pair is a different coordinate f' .

8.4. Sharpness of the two-out-of-three hypothesis

The following example shows that the two-out-of-three hypothesis in Theorem 62 cannot be weakened to membership in a single arrangement. It also shows that a pair derived from a middle nucleus need not itself be a binary nuclear point for the original kernel M . We use the same four-element monoid as in Example 35. Let $\mathcal{C} = \{e, a, c, d\}$, with multiplication table

$\cdot$	e	a	c	d
e	e	a	c	d
a	a	e	c	d
c	c	d	c	d
d	d	c	c	d

and define $p : \mathcal{C} \rightarrow \mathbb{R}$ by

$$p(e) = 0, \quad p(a) = -3, \quad p(c) = 1, \quad p(d) = 4.$$

The table is associative, and e is the unit. Write vectors in the order (e, a, c, d) , and set

$$u = (0, 0, -1, -4), \quad f = (-6, 0, -7, -10), \quad v = (0, 6, 3, 3).$$

The middle coordinate f is the middle closure of the principal seed at a . Put

$$g = N^* f.$$

Direct calculation gives

g	e	a	c	d
e	0	6	3	3
a	6	6	6	6
c	6	6	2	2
d	0	6	-1	-1

Direct calculation gives

$$N_* g = f, \quad g^* u = v, \quad g_* v = u.$$

Hence

$$(f, g) \in \text{Nuc}(N), \quad (u, v) \in \text{Nuc}(g),$$

and therefore

$$(u, f, v) \in T_N.$$

However, the missing middle-coordinate equation gives

$$\left(\inf_{x,z} (M_3(x, b, z) - u(x) - v(z)) \right)_b = (-6, 0, -4, -7),$$

which is not

$$f = (-6, 0, -7, -10).$$

The displayed vector is pointwise larger than f , strictly at c and d . Thus (B_f) fails, and hence

$$(u, f, v) \notin T_{\text{bal}}.$$

Consequently

$$(u, f, v) \in T_N \setminus T_{\text{bal}}.$$

The same example also separates derived nuclearity from binary nuclearity for the original kernel M . Although

$$(u, v) \in \text{Nuc}(g),$$

we have

$$M^*u = (0, 0, 0, 0) \neq (0, 6, 3, 3) = v,$$

and

$$M_*v = (-6, 0, -4, -7) \neq (0, 0, -1, -4) = u.$$

Therefore

$$(u, v) \notin \text{Nuc}(M).$$

9. Product envelopes and factor gaps

In the finite real case the presheaf-coordinate calculus admits a projective bookkeeping of product witnesses. This section records the finite coordinate definitions needed for the product-envelope and factor-gap formulas. The gap quantities below are the invariants that connect the middle product with the polyhedral and metric geometry of projective nuclei developed in the companion paper.

Throughout this section assume that $\mathcal{C}$ is finite and that $p: \mathcal{C} \rightarrow \mathbb{R}$ is real-valued. Thus the ternary measurement N is a finite real matrix. Let $\text{Nuc}_{\mathbb{R}}(N)$ be the locus of points $(f, g) \in \text{Nuc}(N)$ whose presheaf and copresheaf coordinates are finite real-valued functions. Define the projective nucleus used in this section by

$$\mathbb{P}\text{Nuc}(N) := \text{Nuc}_{\mathbb{R}}(N) / \sim,$$

where

$$(f, g) \sim (f + \lambda, g - \lambda), \quad \lambda \in \mathbb{R}.$$

For $X = (f, g) \in \text{Nuc}_{\mathbb{R}}(N)$, write $[X]$ for its projective class. We keep the standard nuclear coordinates. Let

$$X = (f, g), \quad Y = (f', g')$$

be points of $\text{Nuc}_{\mathbb{R}}(N)$, so

$$g = N^*f, \quad g' = N^*f'.$$

Define

$$M_4(a, b, c, d) := p(abcd) - p(a) - p(b) - p(c) - p(d).$$

The product

$$Z := X \odot Y = (f'', g'')$$

is represented by the nuclear pair whose copresheaf coordinate is

$$g''(x, z) = \min_{b, b' \in \mathcal{C}} (M_4(x, b, b', z) - f(b) - f'(b')), \quad (9.1)$$

and whose presheaf coordinate is recovered by

$$f''(y) = N_* g''(y) = \min_{x, z \in \mathcal{C}} (M_3(x, y, z) - g''(x, z)). \quad (9.2)$$

Equivalently, g'' is the boundary envelope obtained by minimizing over the two middle factors, and f'' is the middle closure recovered from that envelope.

COROLLARY 63 Projective product envelope. *The assignment*

$$\mu_{\odot}: \mathbb{PNuc}(N) \times \mathbb{PNuc}(N) \longrightarrow \mathbb{PNuc}(N), \quad ([X], [Y]) \longmapsto [Z]$$

is well-defined. More explicitly, if

$$X = (f, g) \quad \text{is replaced by} \quad X_{\lambda} = (f + \lambda, g - \lambda)$$

and

$$Y = (f', g') \quad \text{is replaced by} \quad Y_{\mu} = (f' + \mu, g' - \mu),$$

then the output coordinates transform as

$$g'' \mapsto g'' - \lambda - \mu, \quad f'' \mapsto f'' + \lambda + \mu.$$

Consequently the product gap

$$\Delta_{X,Y}(y; x, z) := M_3(x, y, z) - f''(y) - g''(x, z) \quad (9.3)$$

is independent of the chosen affine representatives of $[X]$ and $[Y]$.

Proof. Formula (9.1) is the middle-product convolution formula in the finite real-valued case. Replacing f by $f + \lambda$ and f' by $f' + \mu$ subtracts $\lambda + \mu$ from every term in the minimum defining g'' , hence sends g'' to $g'' - \lambda - \mu$. Applying N_* then sends f'' to $f'' + \lambda + \mu$. Thus the projective class $[Z]$ is independent of the representatives. The same two shifts cancel in $M_3(x, y, z) - f''(y) - g''(x, z)$, proving the invariance of $\Delta_{X,Y}$. $\square$

9.1. Factor gaps

The gap $\Delta_{X,Y}$ is the ordinary middle gap of the output point $Z = (f'', g'')$. To see how this output gap is built from the two input points, we also keep track of the factorisations that compute the envelope g'' . For $b, b', x, z \in \mathcal{C}$, define the factor gap

$$\kappa_{X,Y}^{b,b'}(x, z) := M_4(x, b, b', z) - f(b) - f'(b') - g''(x, z). \quad (9.4)$$

Then $\kappa_{X,Y}^{b,b'}(x, z) \geq 0$, and it vanishes exactly when (b, b') realizes the minimum in (9.1) at (x, z) . The factor gap decomposes into input witness gaps and outer minimization slacks. Write

$$\delta_X(b; x, z) := M_3(x, b, z) - f(b) - g(x, z), \quad (9.5)$$

$$\delta_Y(b; x, z) := M_3(x, b, z) - f'(b) - g'(x, z). \quad (9.6)$$

These are the ordinary middle gap matrices of the input points $X = (f, g)$ and $Y = (f', g')$. Define also

$$\sigma_{X,Y}^L(b'; x, z) := g(x, b'z) + M(b', z) - f'(b') - g''(x, z), \quad (9.7)$$

$$\sigma_{X,Y}^R(b; x, z) := M(x, b) - f(b) + g'(xb, z) - g''(x, z). \quad (9.8)$$

The nonnegativity of these two slacks follows from the equivalent one-sided forms of the envelope:

$$g''(x, z) = \min_{b' \in \mathcal{C}} (g(x, b'z) + M(b', z) - f'(b')), \quad (9.9)$$

$$g''(x, z) = \min_{b \in \mathcal{C}} (M(x, b) - f(b) + g'(xb, z)). \quad (9.10)$$

PROPOSITION 64 Factorisation of convolution gaps. *For all $x, z, b, b' \in \mathcal{C}$,*

$$\kappa_{X,Y}^{b,b'}(x, z) = \delta_X(b; x, b'z) + \sigma_{X,Y}^L(b'; x, z) = \delta_Y(b'; xb, z) + \sigma_{X,Y}^R(b; x, z). \quad (9.11)$$

Consequently, for fixed x, z, b, b' , the following conditions are equivalent:

- (i) (b, b') computes the envelope $g''(x, z)$, equivalently $\kappa_{X,Y}^{b,b'}(x, z) = 0$;
- (ii) $\delta_X(b; x, b'z) = 0$ and $\sigma_{X,Y}^L(b'; x, z) = 0$;
- (iii) $\delta_Y(b'; xb, z) = 0$ and $\sigma_{X,Y}^R(b; x, z) = 0$.

In particular, every envelope-computing factorisation satisfies

$$\delta_X(b; x, b'z) = \delta_Y(b'; xb, z) = \sigma_{X,Y}^L(b'; x, z) = \sigma_{X,Y}^R(b; x, z) = 0.$$

Proof. The adjacent splitting identities for M_4 give

$$M_4(x, b, b', z) = M_3(x, b, b'z) + M(b', z)$$

and

$$M_4(x, b, b', z) = M(x, b) + M_3(xb, b', z).$$

Using the first identity,

$$\begin{aligned} \kappa_{X,Y}^{b,b'}(x, z) &= M_3(x, b, b'z) + M(b', z) - f(b) - f'(b') - g''(x, z) \\ &= (M_3(x, b, b'z) - f(b) - g(x, b'z)) \\ &\quad + (g(x, b'z) + M(b', z) - f'(b') - g''(x, z)) \\ &= \delta_X(b; x, b'z) + \sigma_{X,Y}^L(b'; x, z). \end{aligned}$$

The second equality in (9.11) is obtained in the same way from the second adjacent splitting identity. All summands are nonnegative: δ_X and δ_Y are ordinary middle gaps, while σ^L and σ^R are nonnegative by (9.9) and (9.10). Therefore $\kappa_{X,Y}^{b,b'}(x, z) = 0$ holds exactly when the two summands in either displayed decomposition vanish. $\square$

Thus a product witness is not just a minimizing pair (b, b') . It is a pair for which b is visible in the witness geometry of $X = (f, g)$ from the shifted boundary $(x, b'z)$, b' is visible in the witness geometry of $Y = (f', g')$ from the shifted boundary (xb, z) , and the two outer envelope minimizations are sharp. The product gap $\Delta_{X,Y}$ then records the ordinary middle gap of the resulting output point $Z = (f'', g'')$.

These identities are the promised local bridge between the Lambek product and the geometry of execution. They identify the witness data organized by the chamber theory in the companion paper: the product witness, the two shifted input witnesses, the outer envelope sharpness conditions, and the output gap.

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